Dragster

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Do you know which cars are called dragsters?
What features do they have?

Dragsters

Dragsters are special cars that are designed for racing at very short distances - as little as a quarter of a mile. And we can make them electric!



Your dragster will have rear drive wheels powered by two motors. The front wheels don't have the steering mechanics and allow only straight movement.



Find:

- smarthub
- motors
- front wheels
- rear wheels



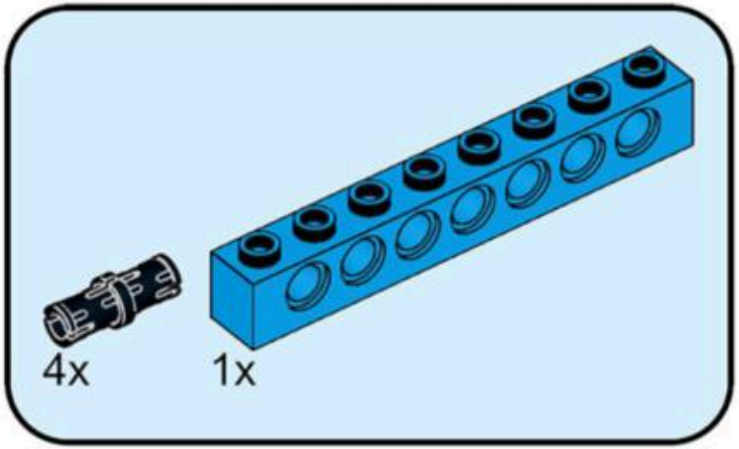
The robot has some aerodynamic elements. The front and rear wings are used to produce the downforce. This force keeps the dragster on track and provides the friction to control the car.



Find:

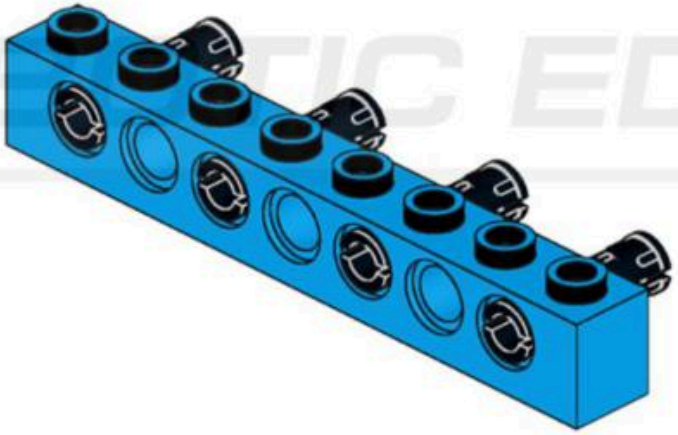
- front wing
- windshield
- rear wing

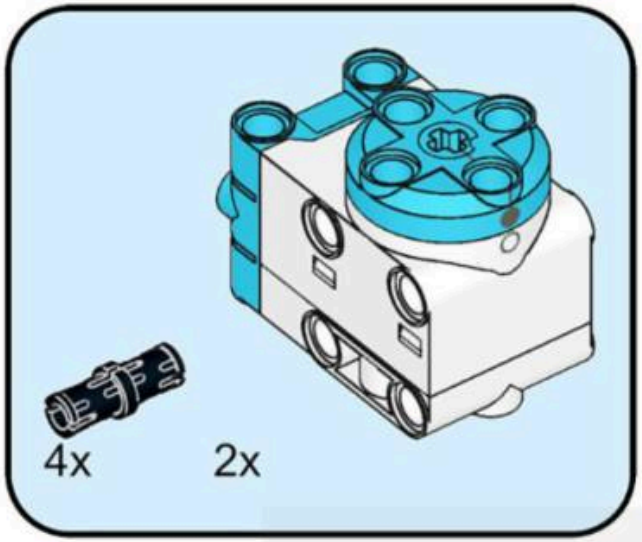




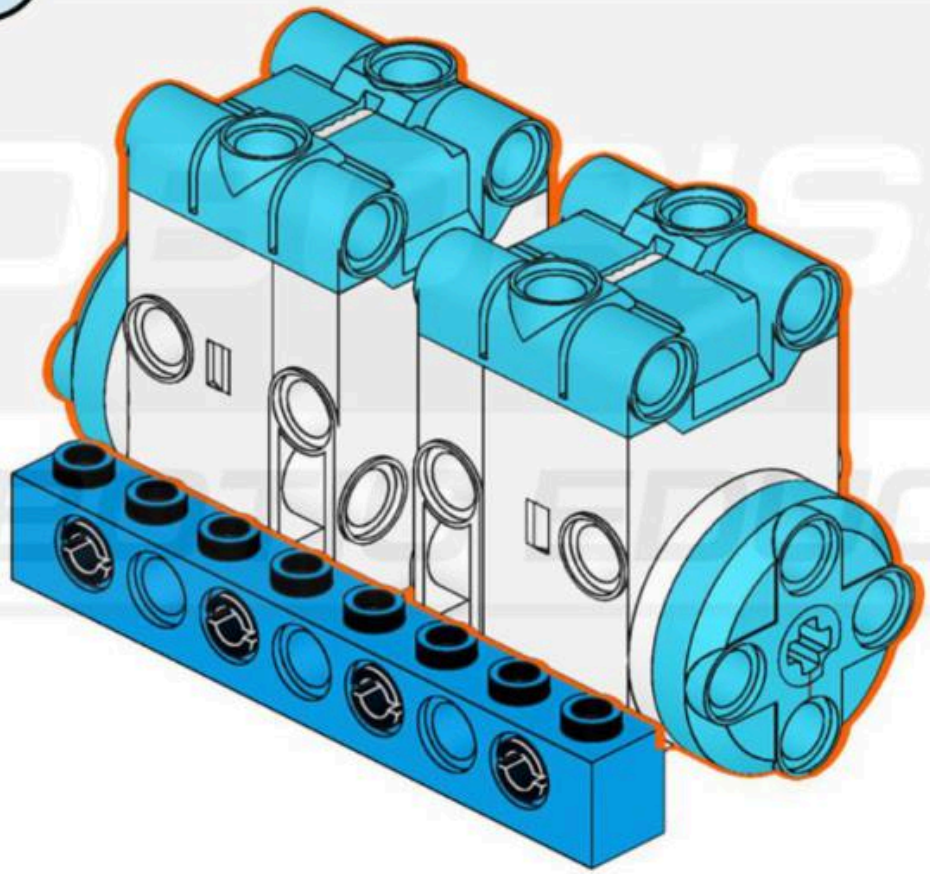
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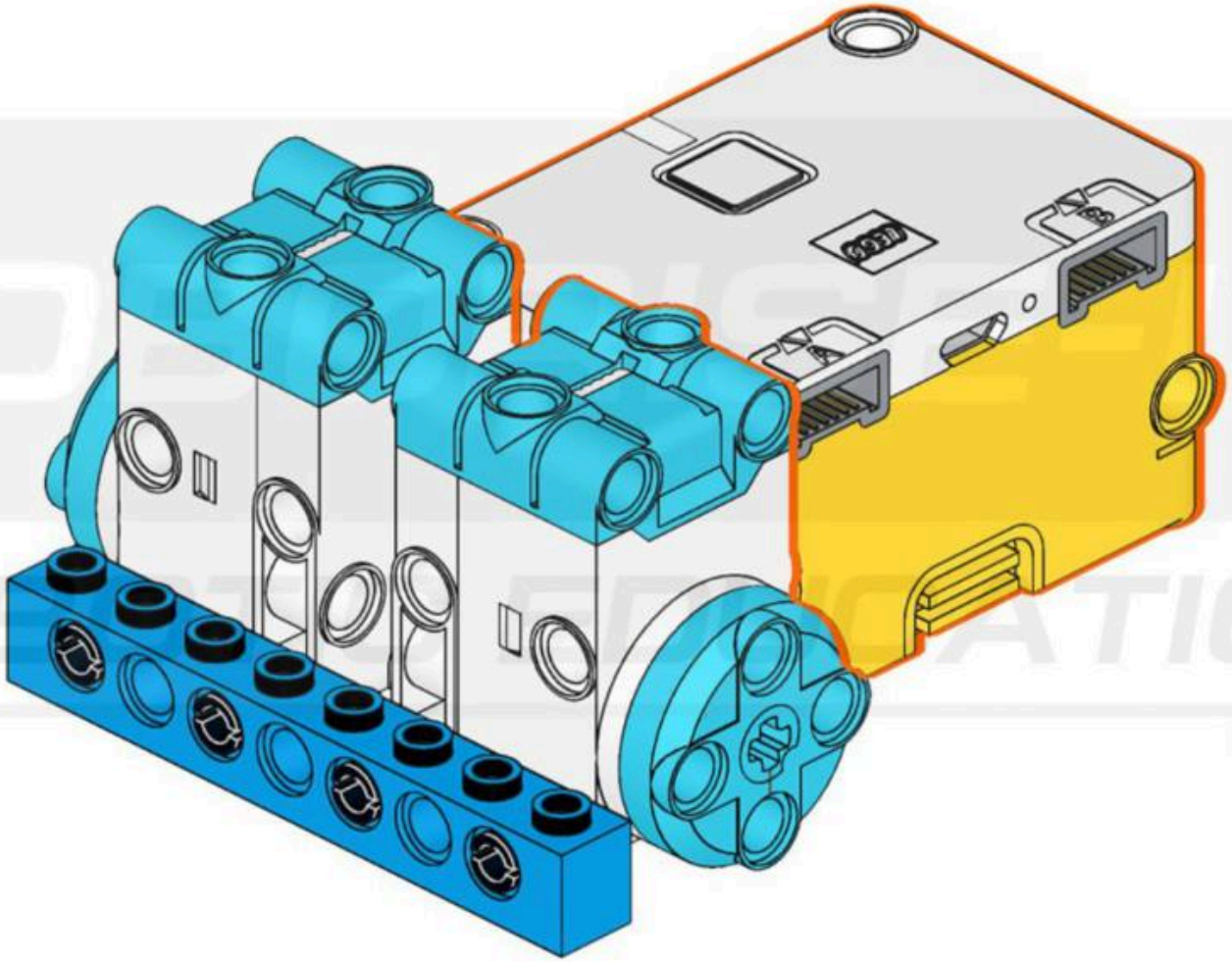




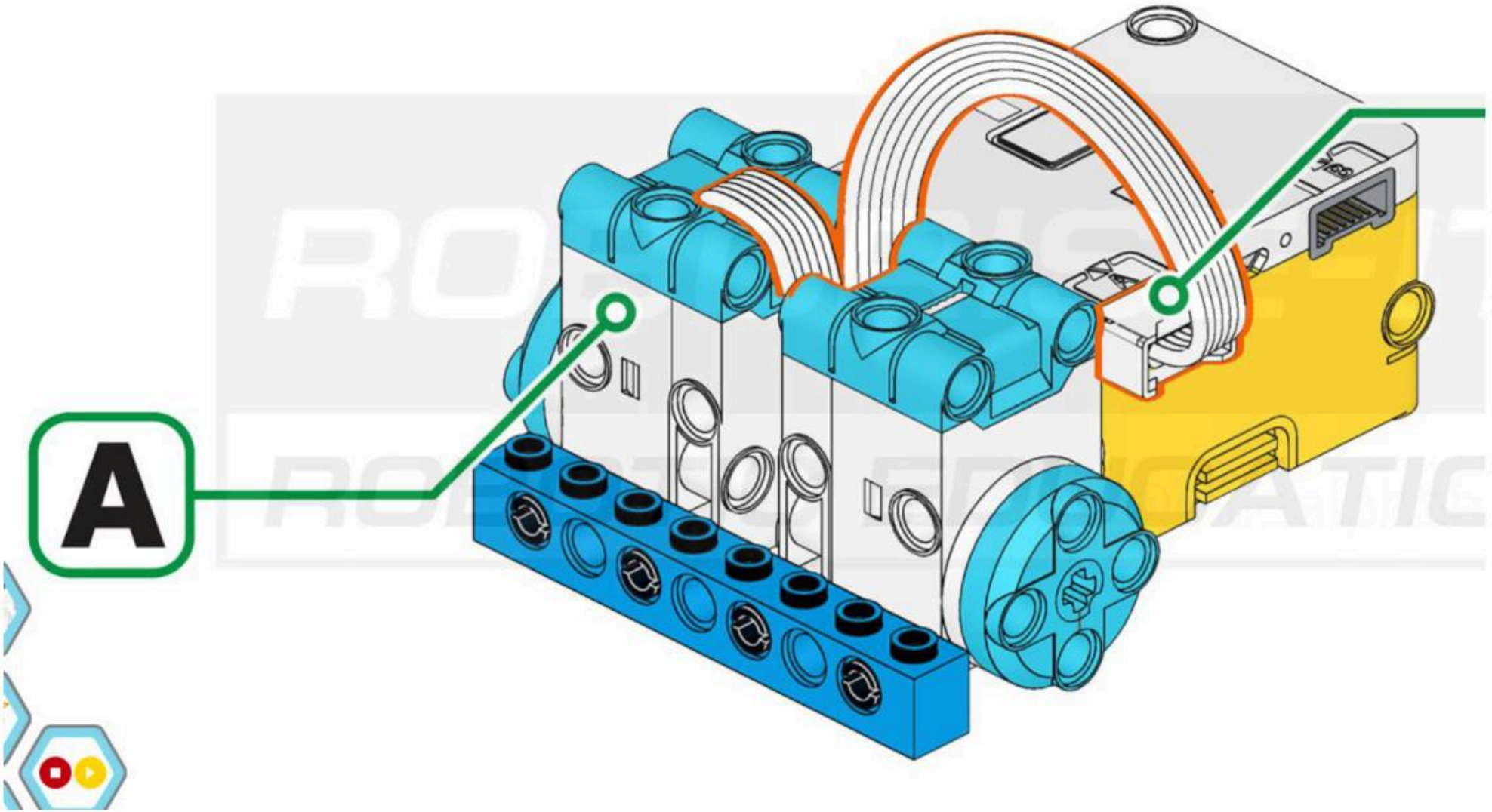
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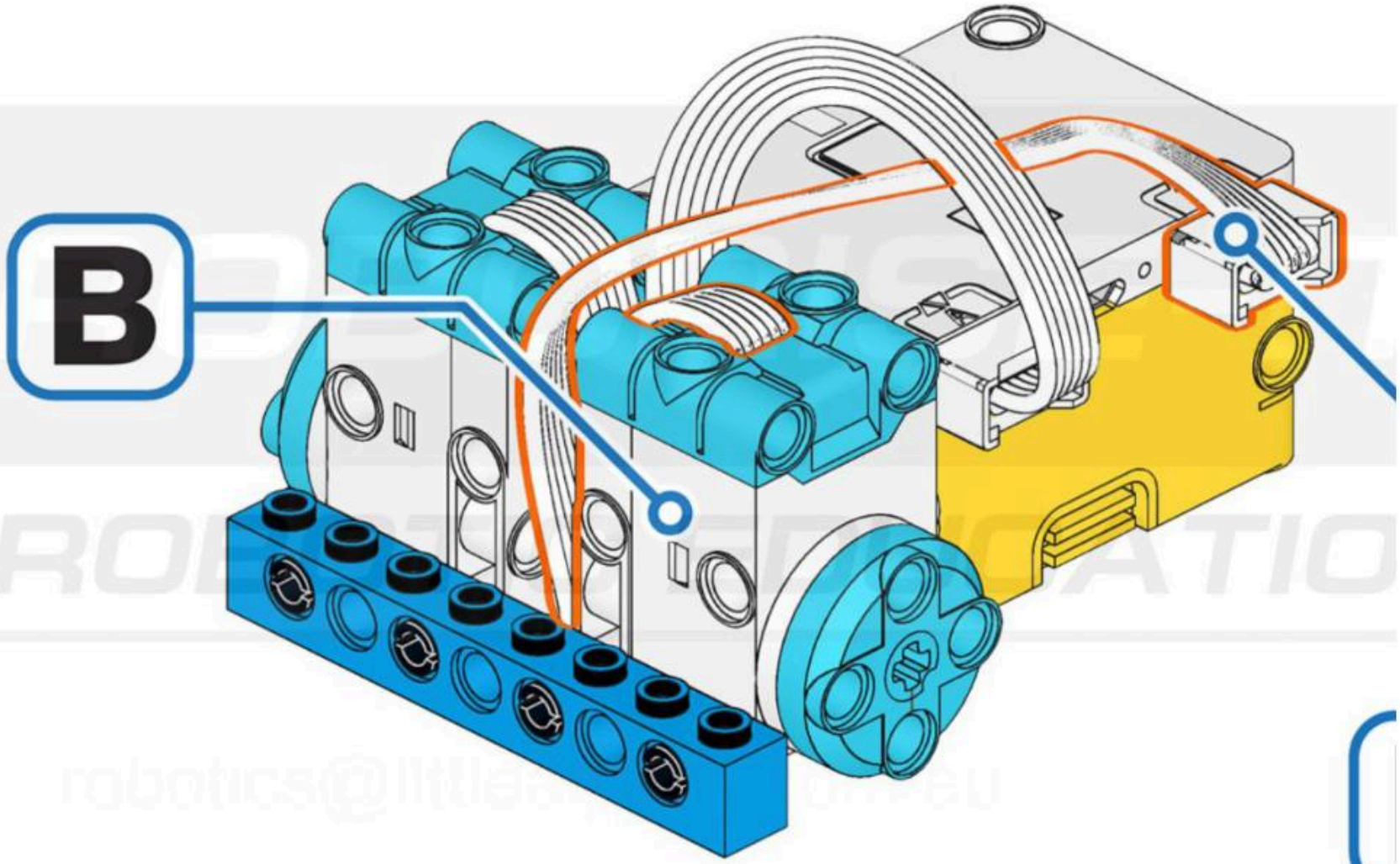
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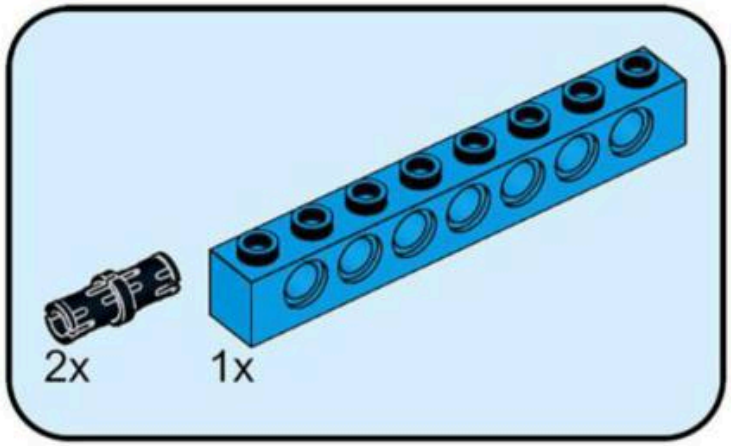


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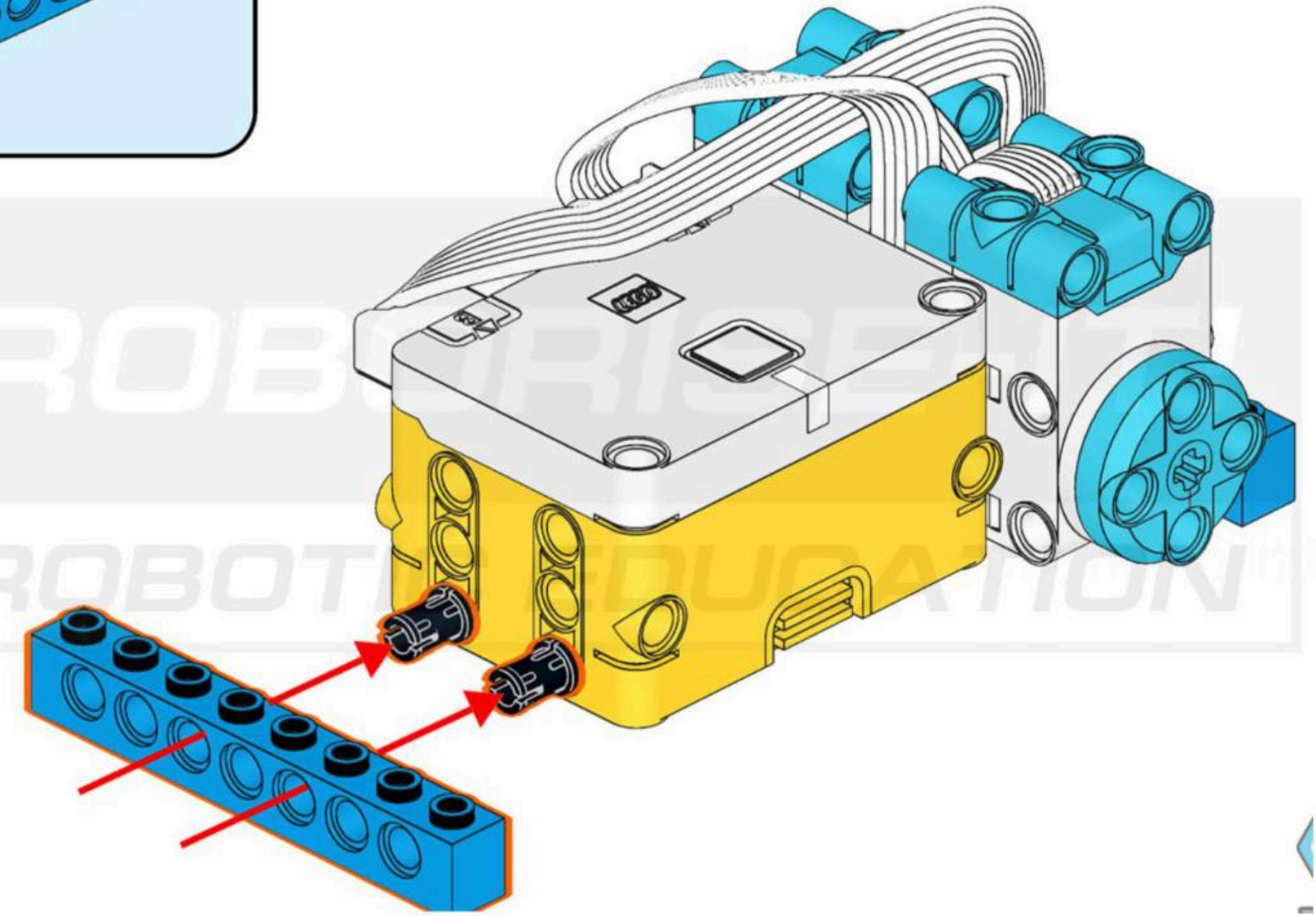


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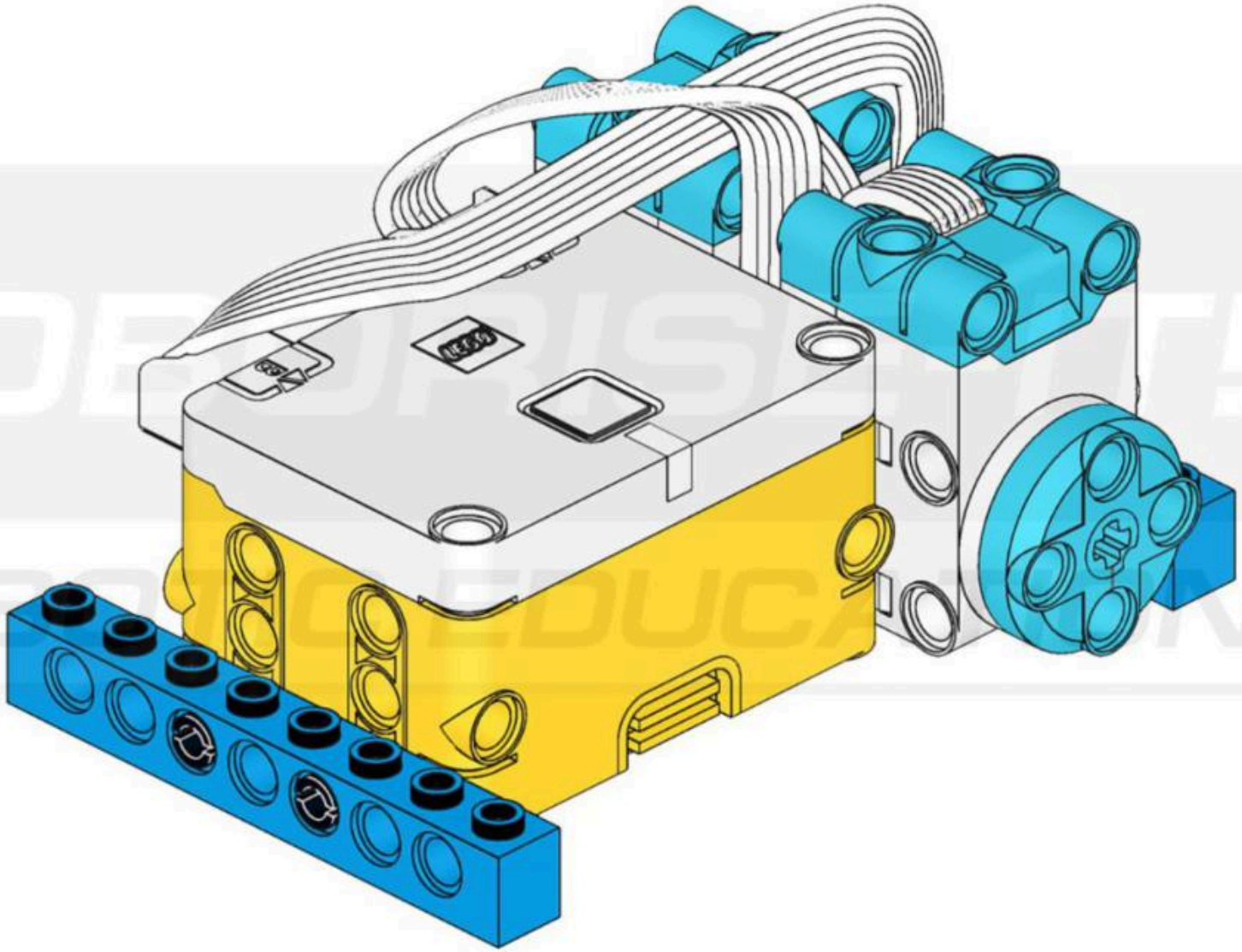




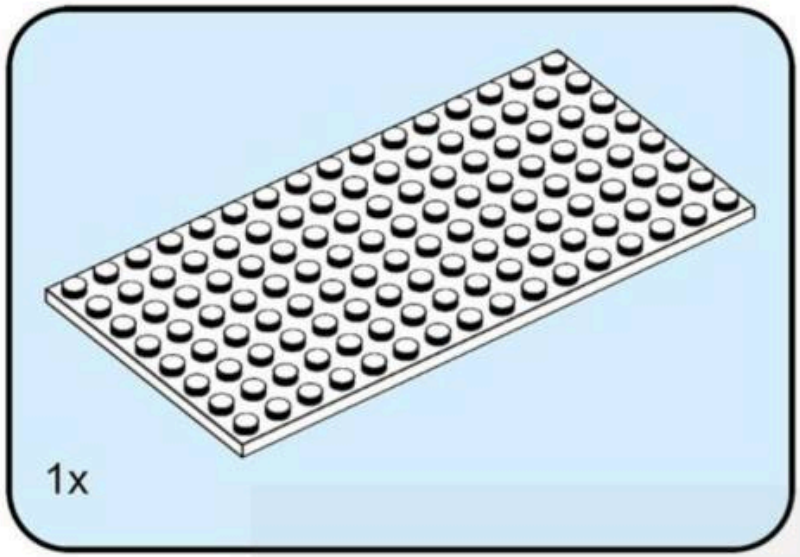
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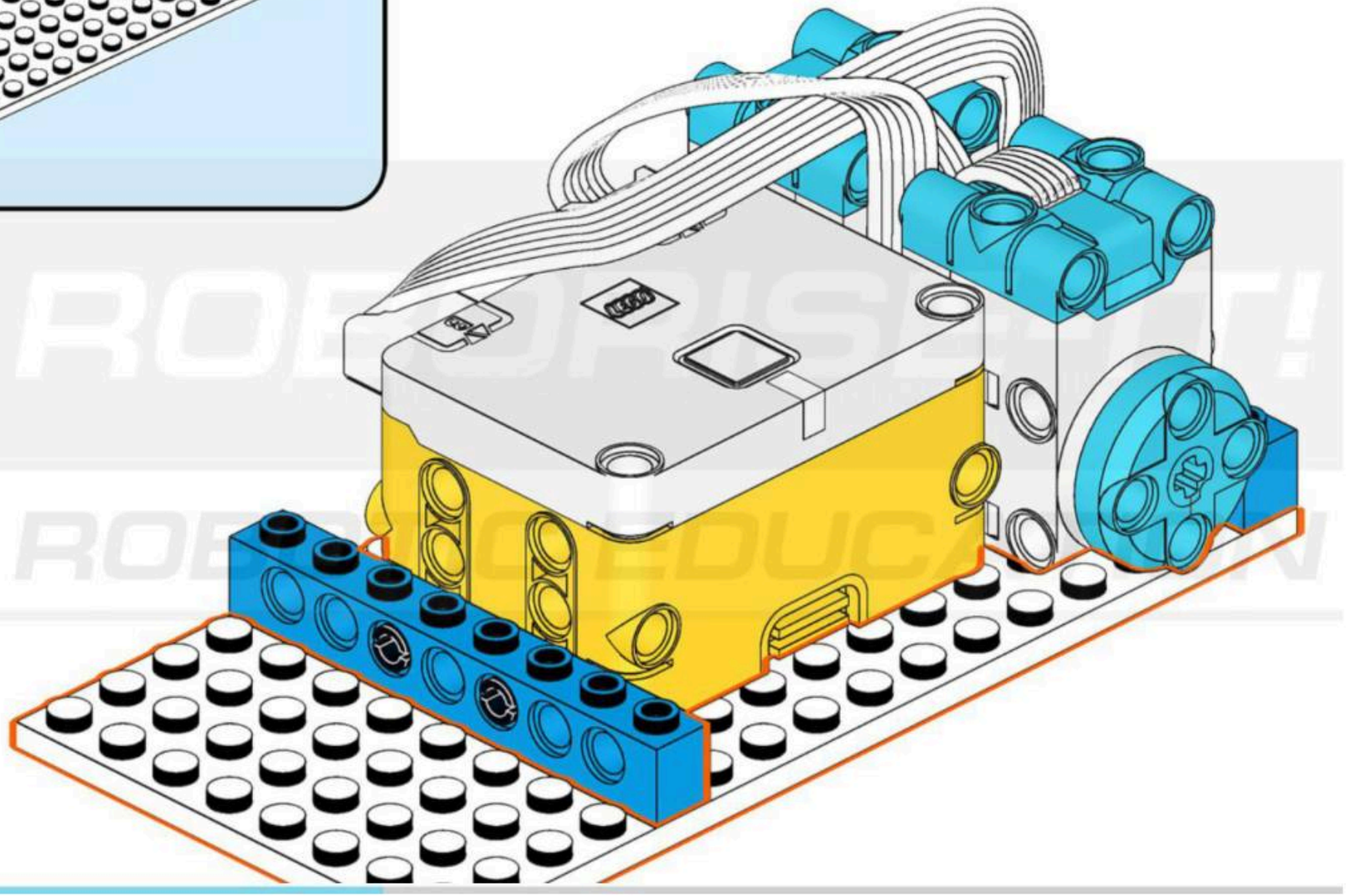
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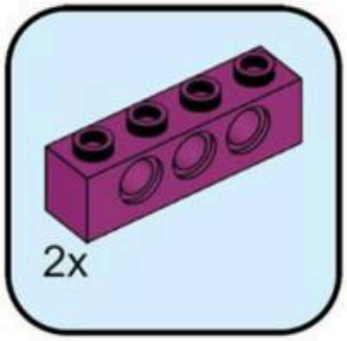


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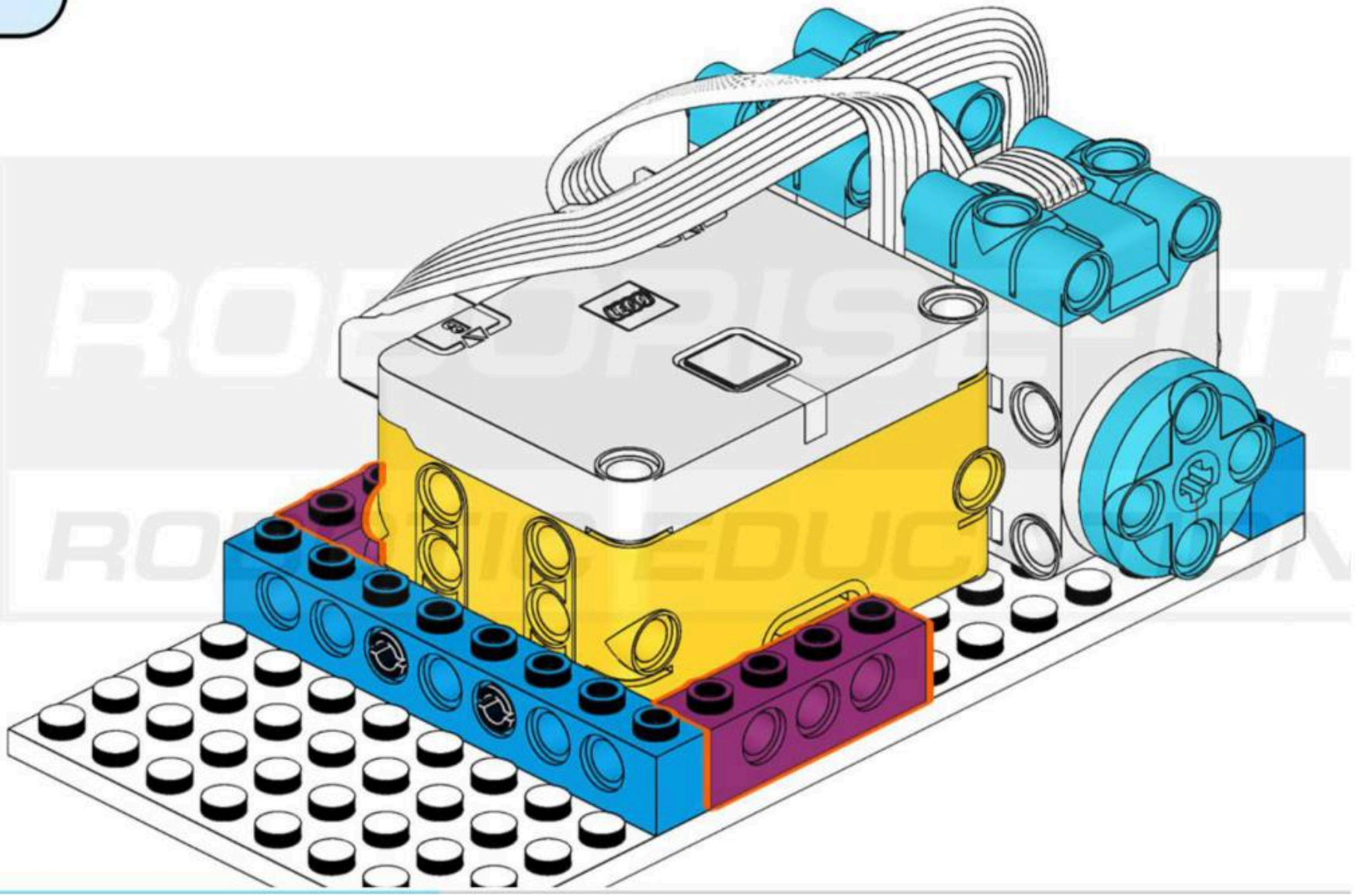


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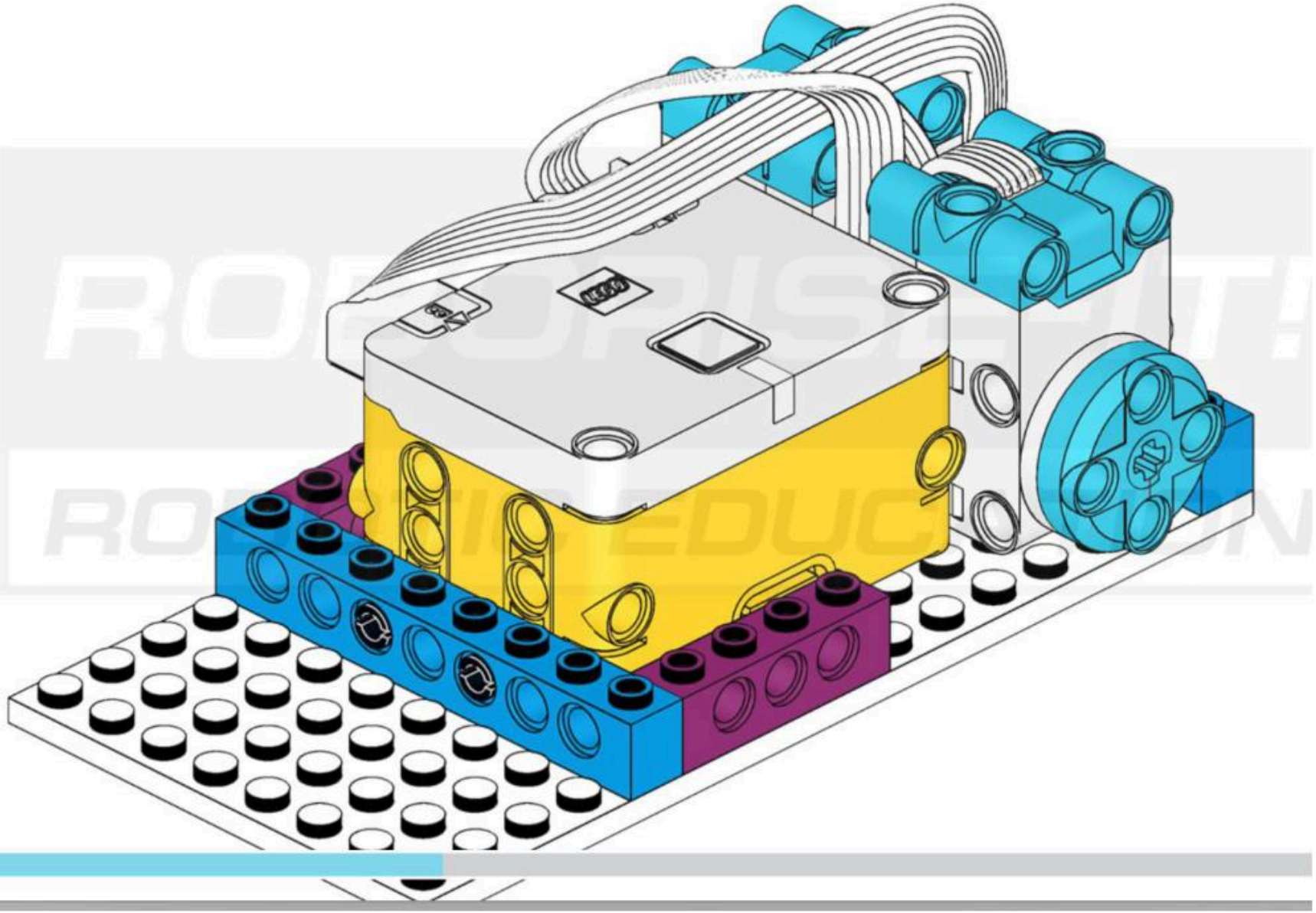


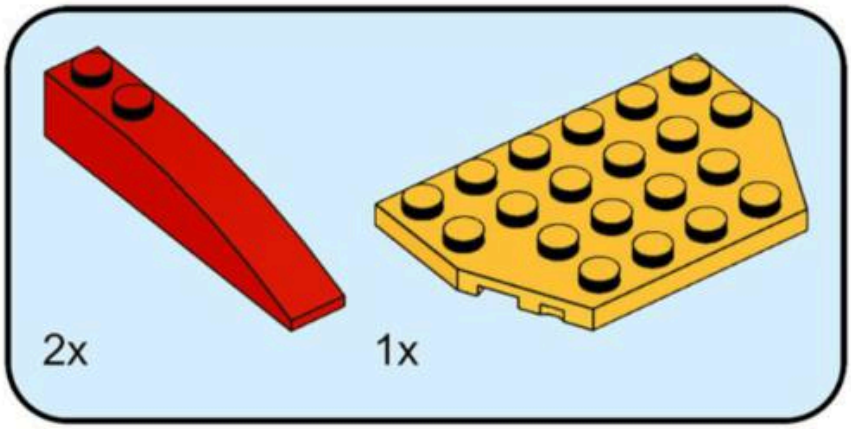


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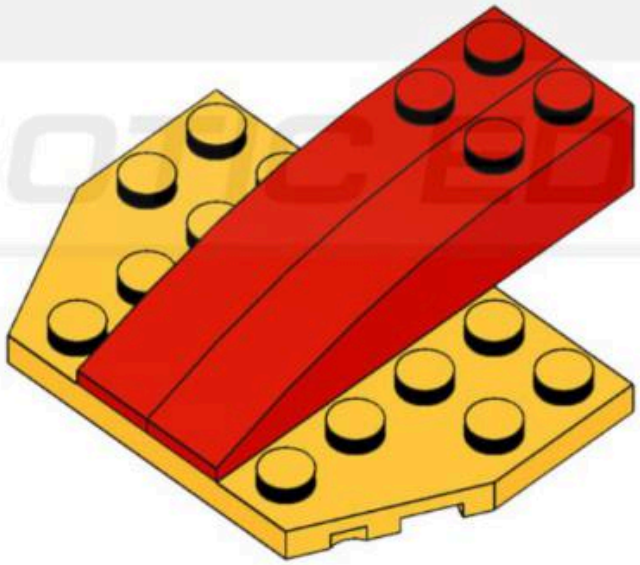


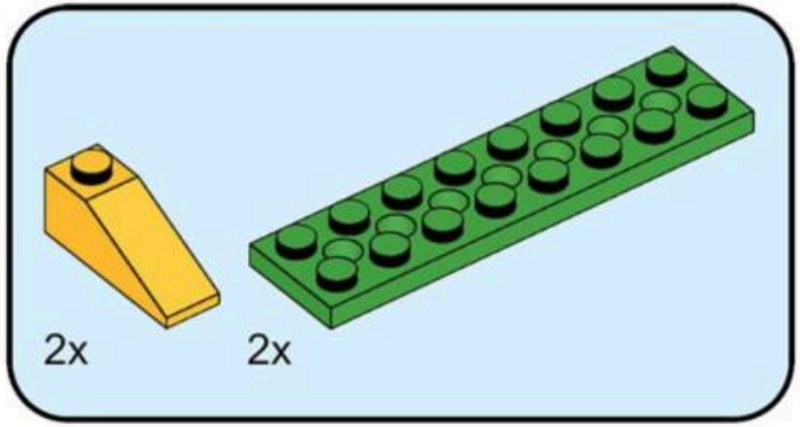
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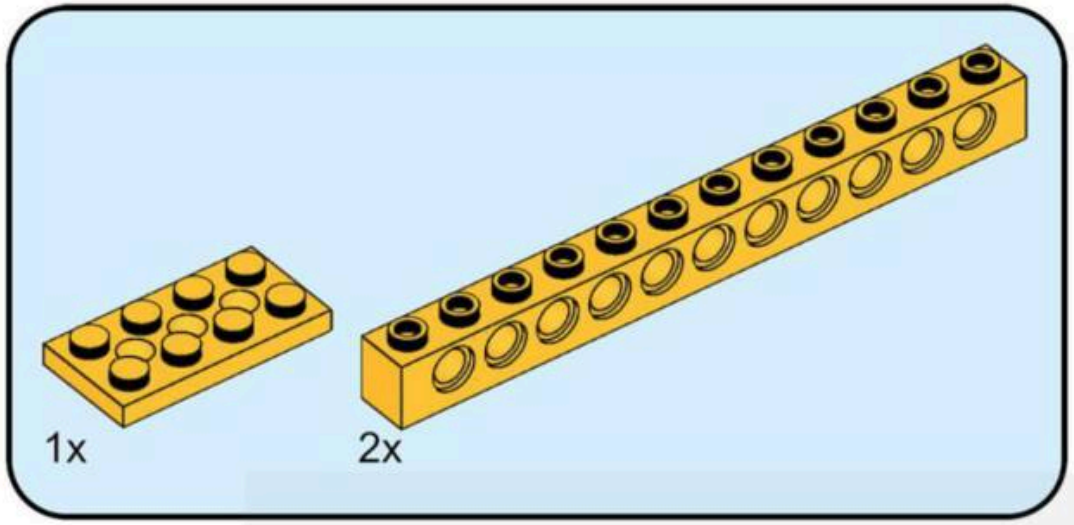




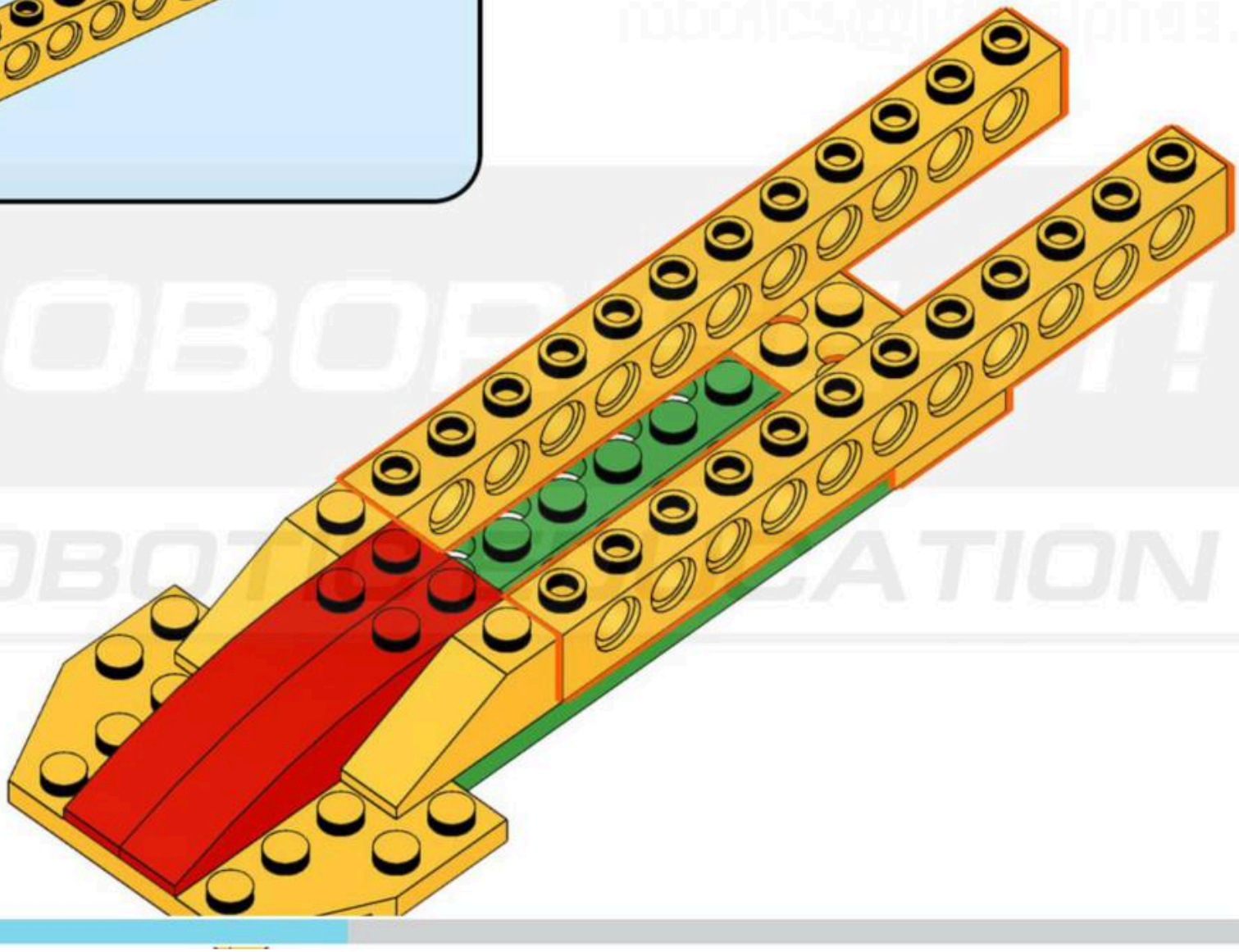
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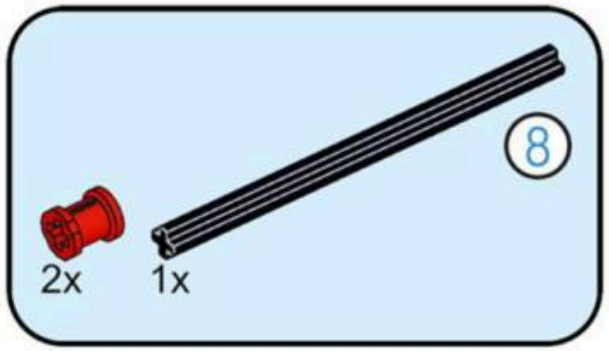


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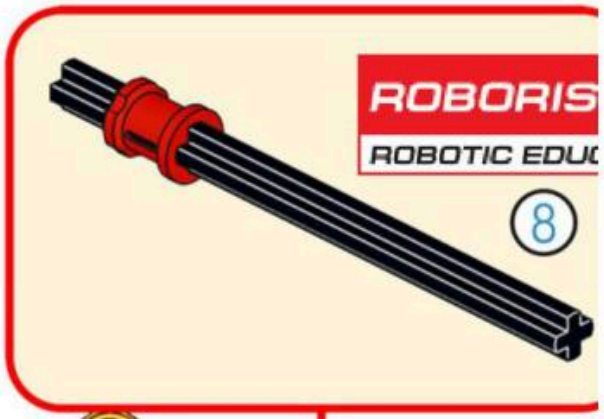


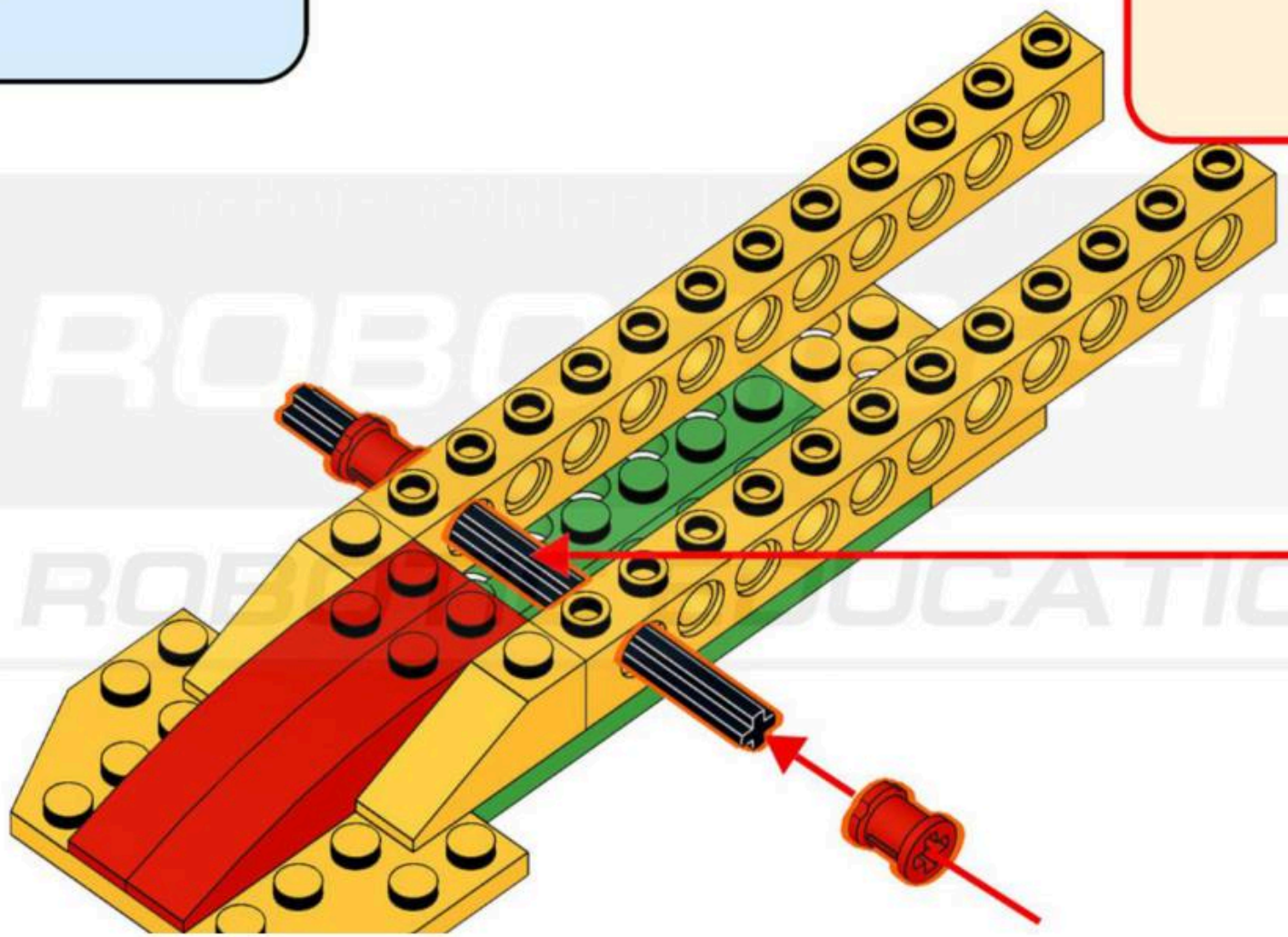
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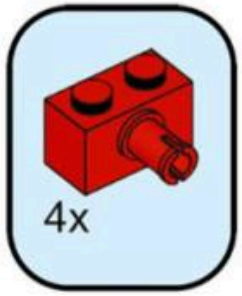


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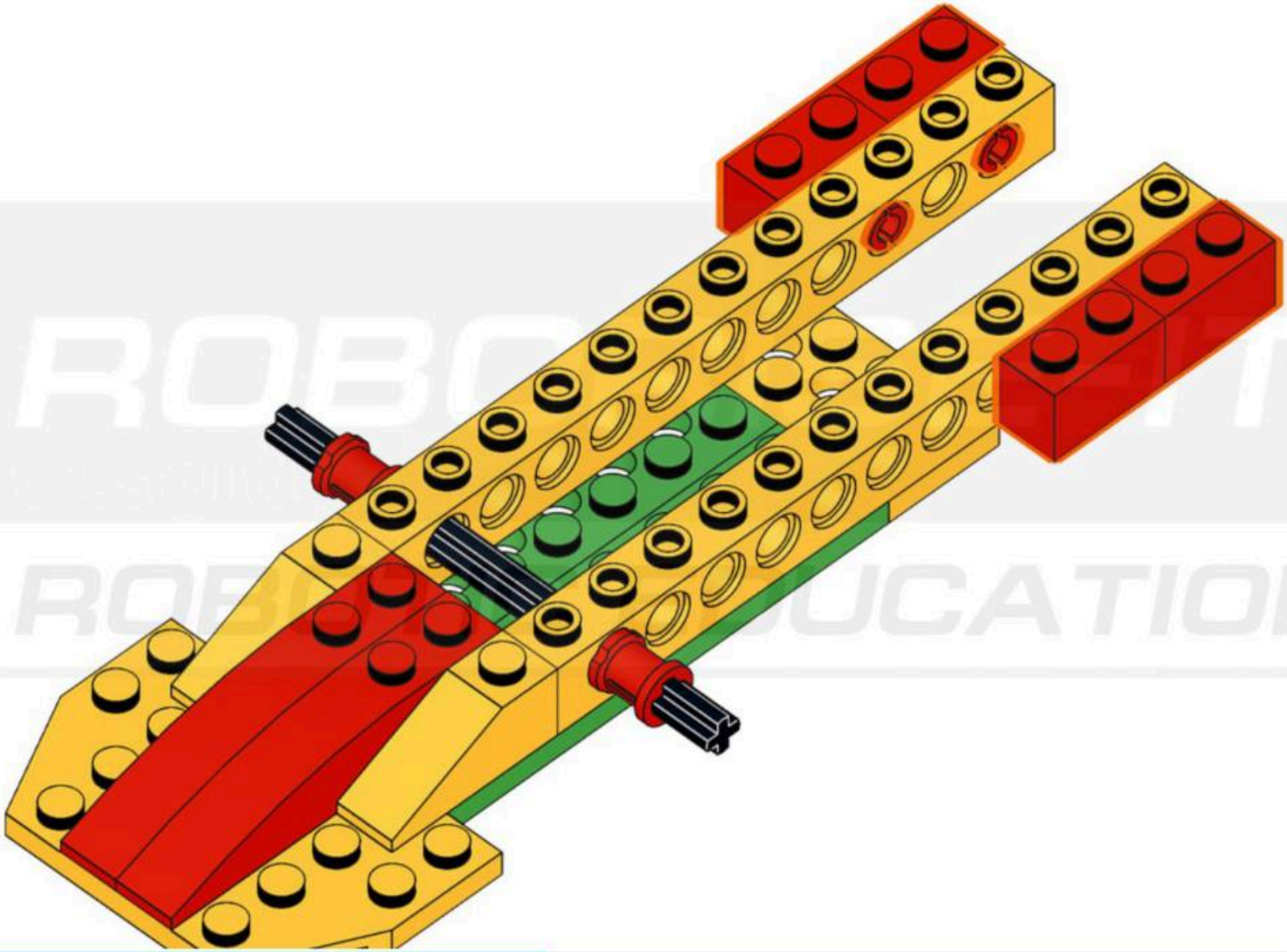


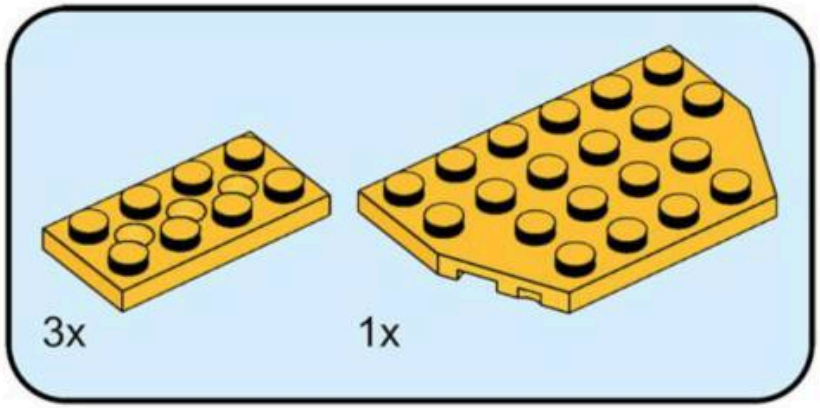


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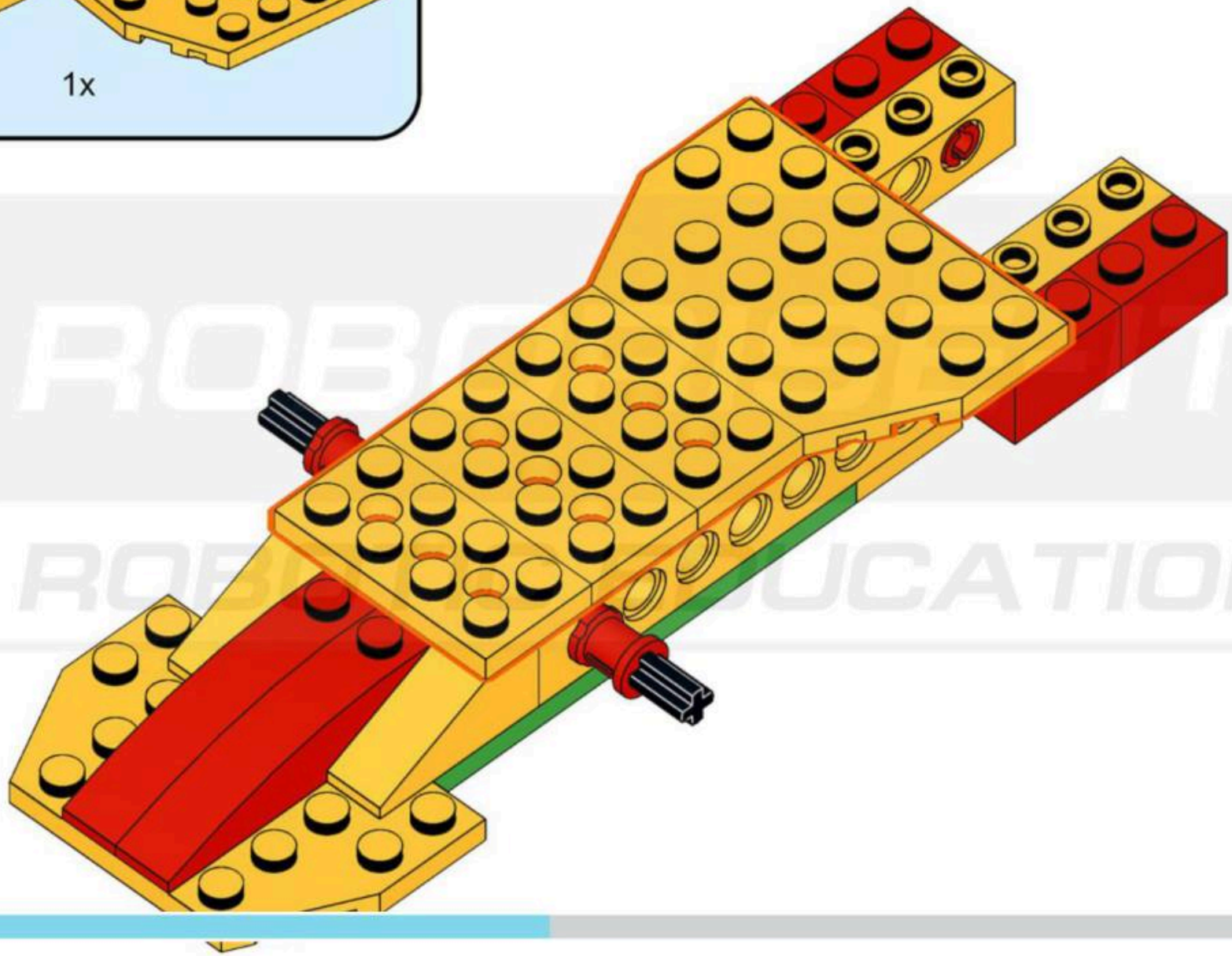


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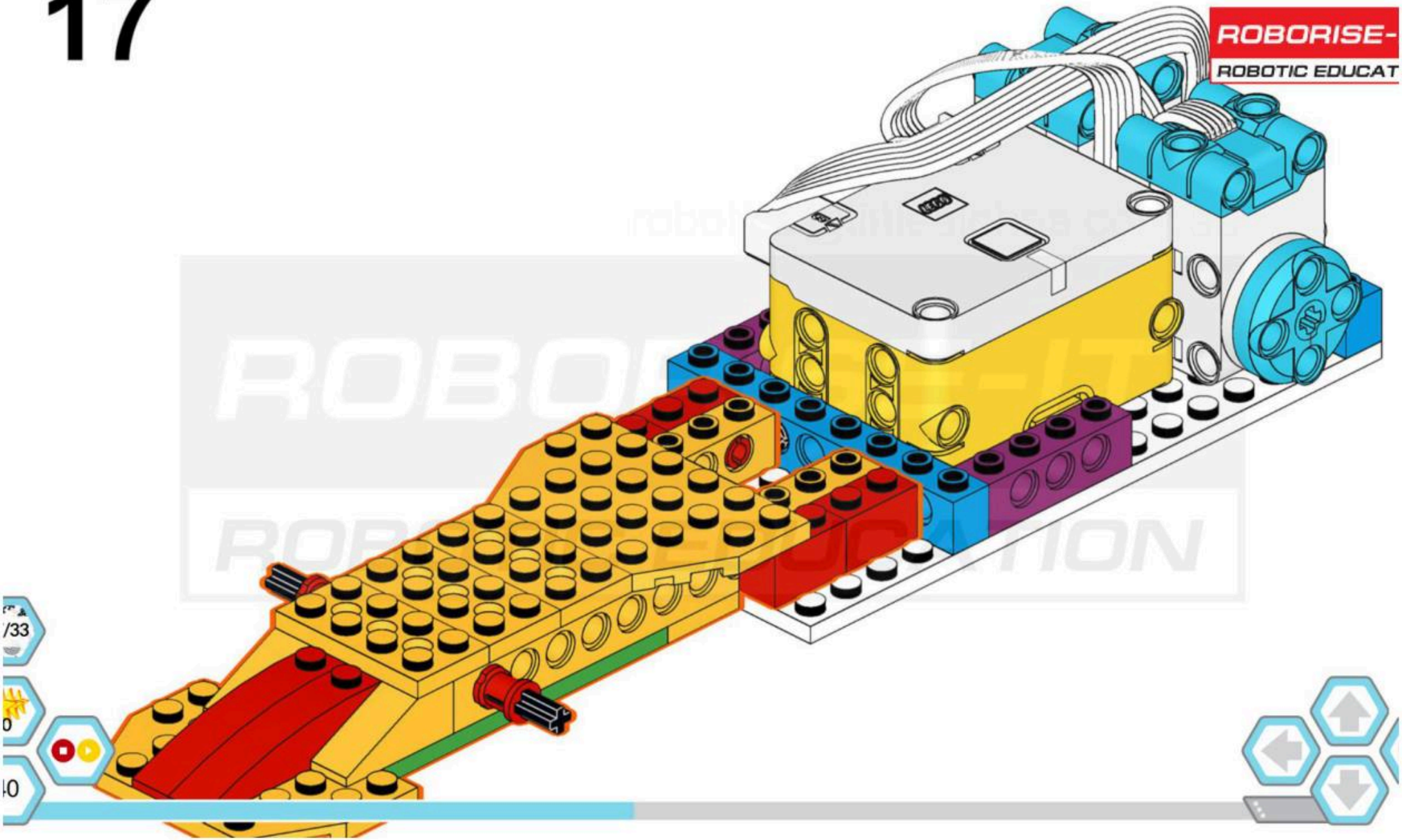


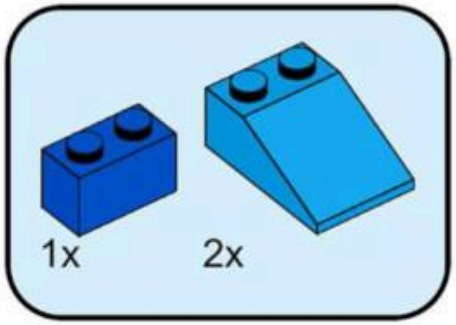


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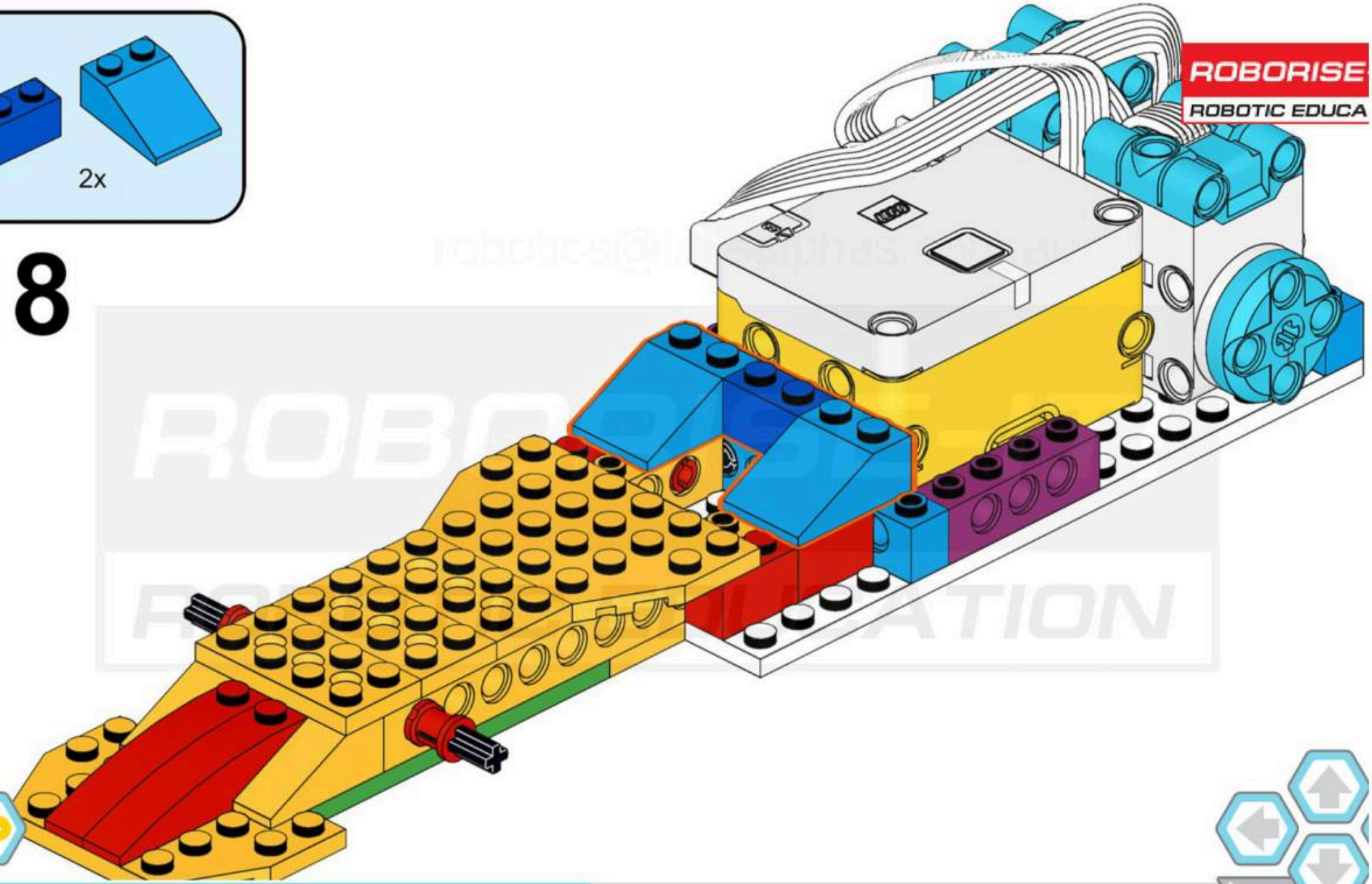


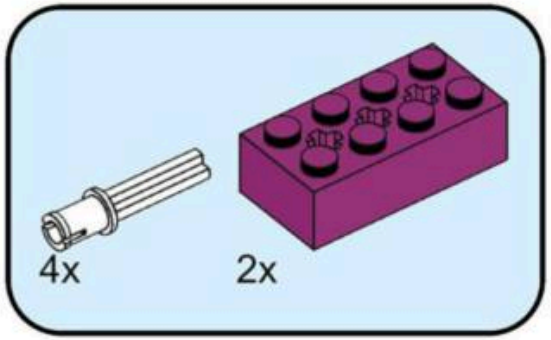
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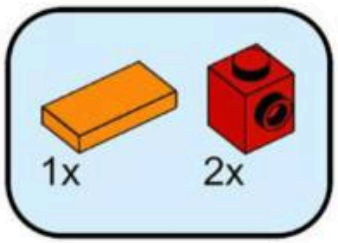




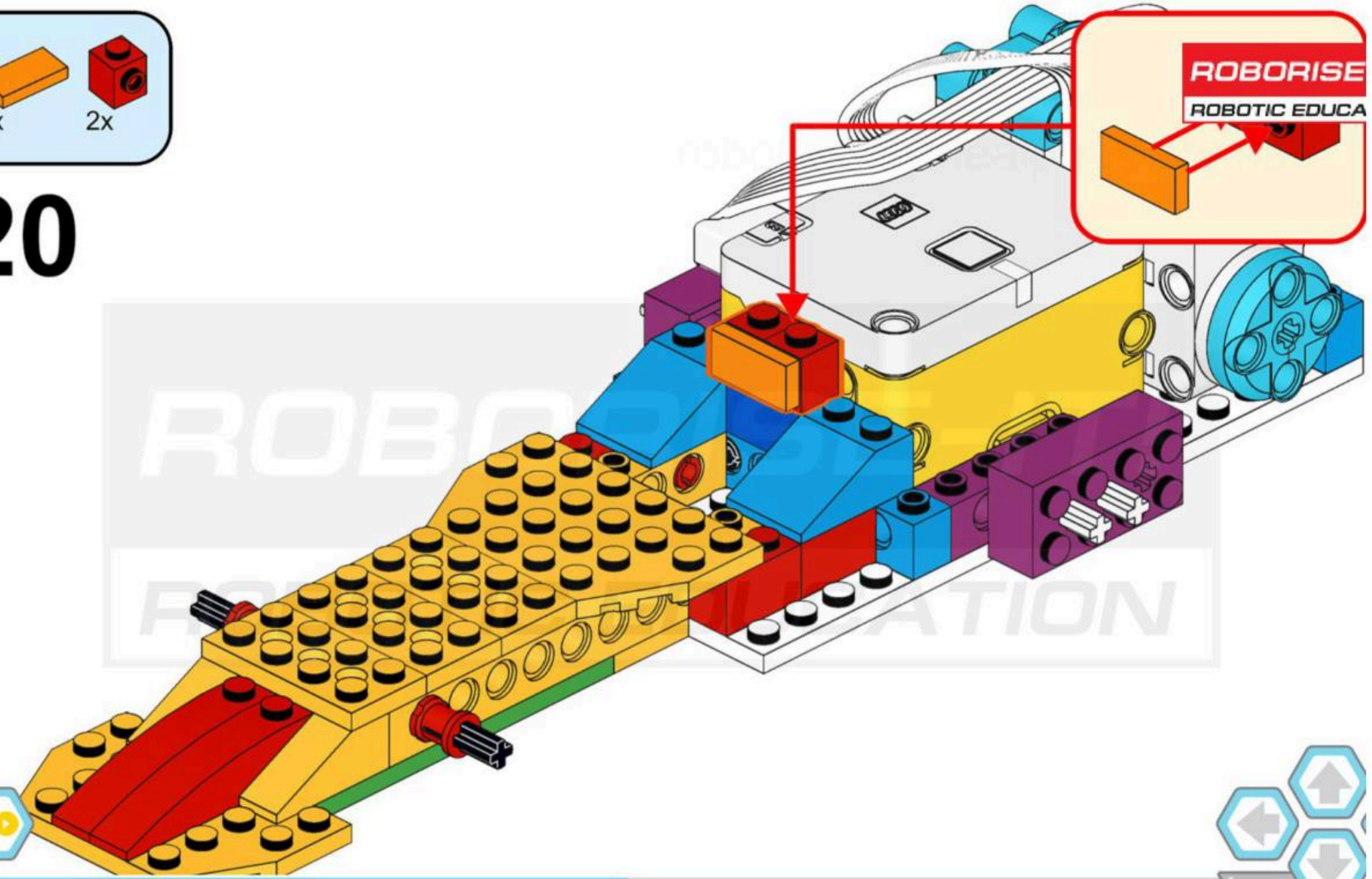
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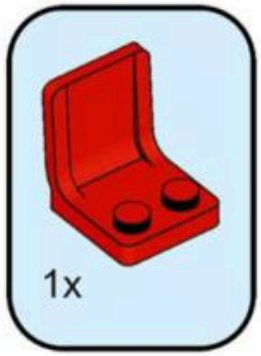
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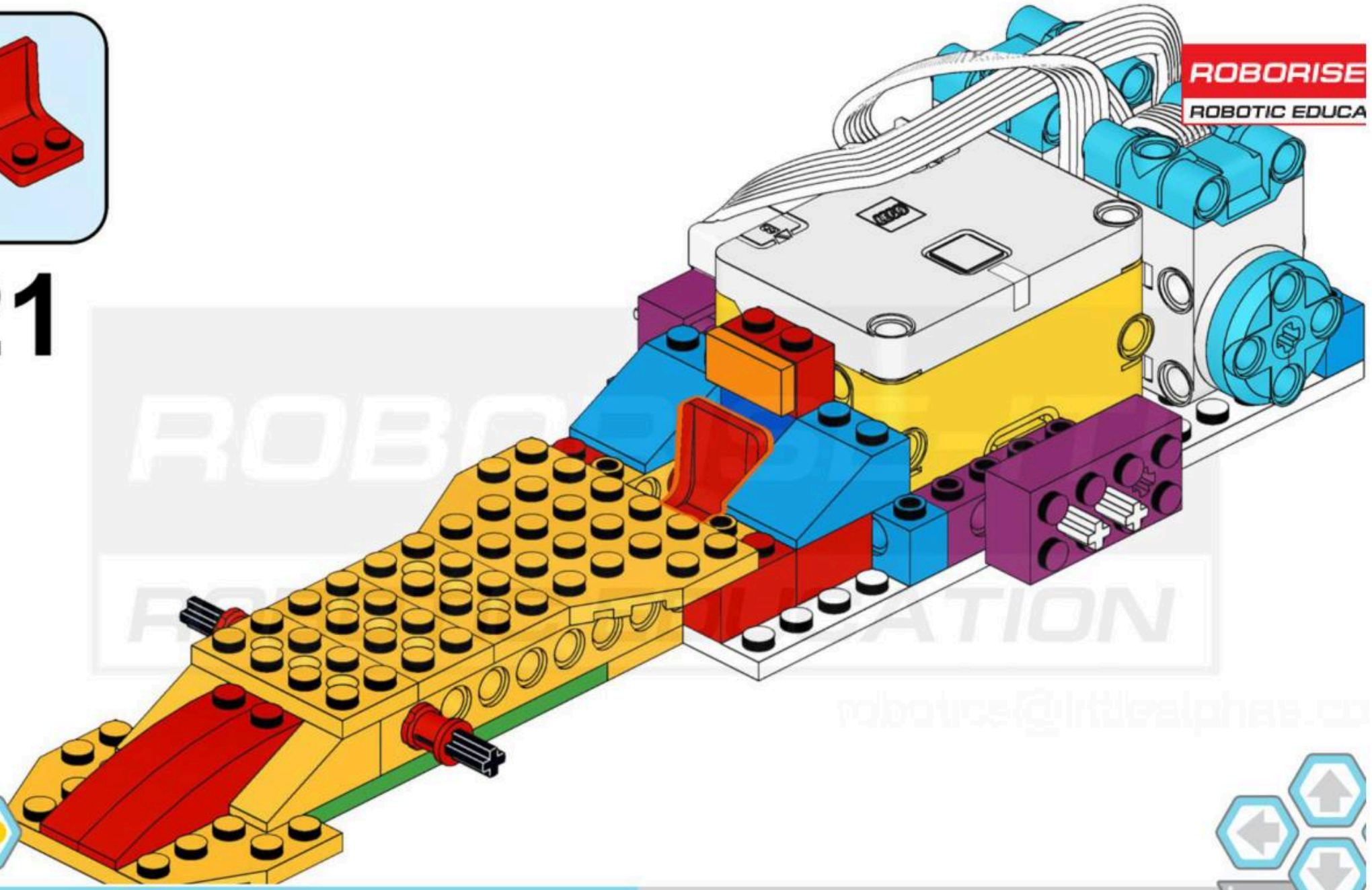


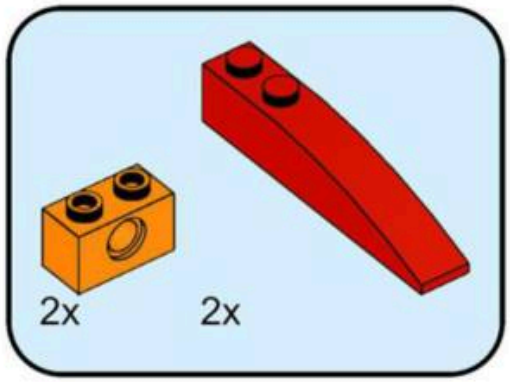
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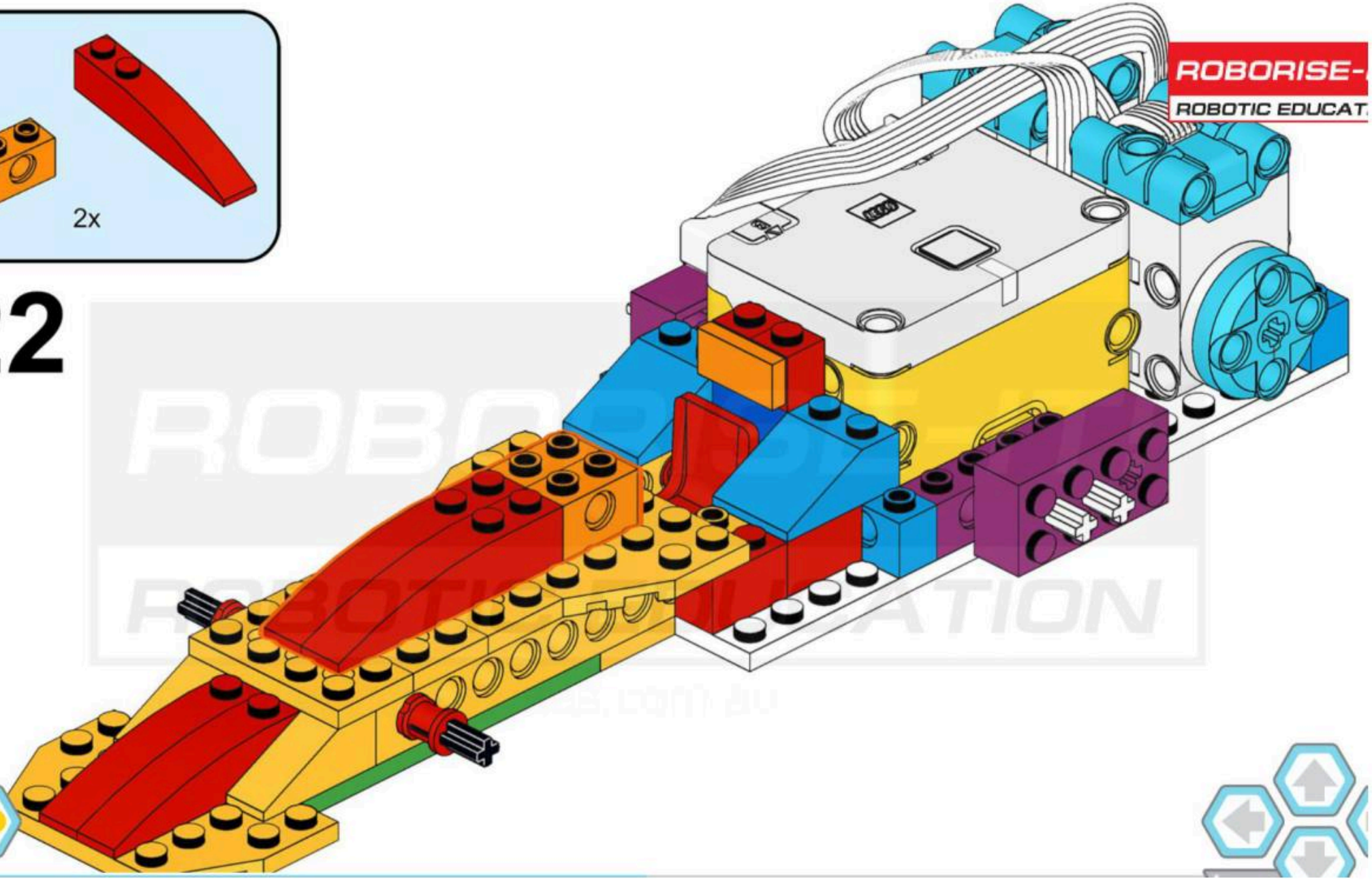


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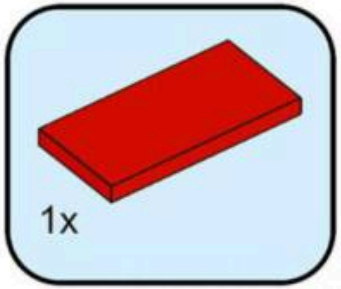




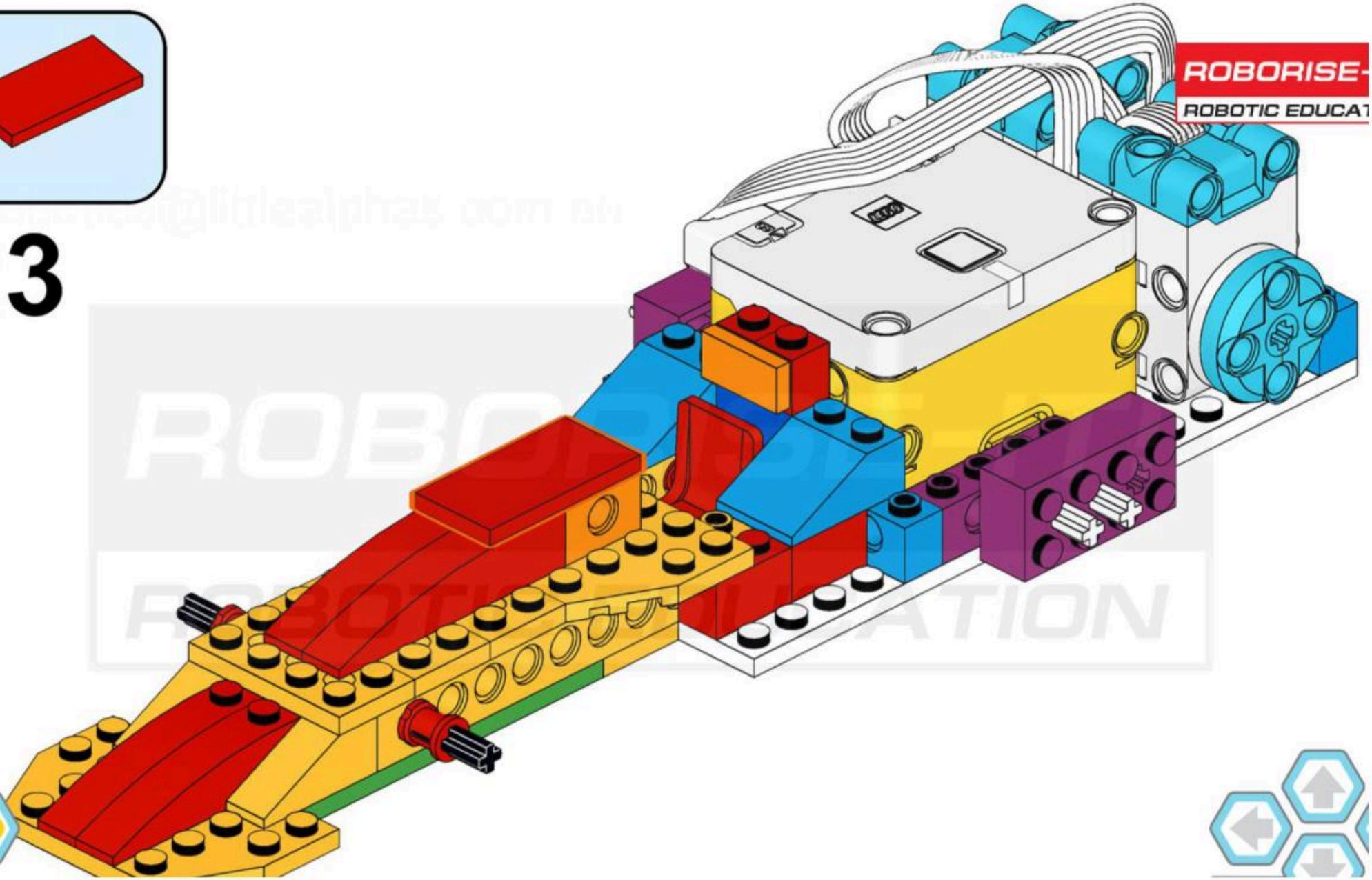
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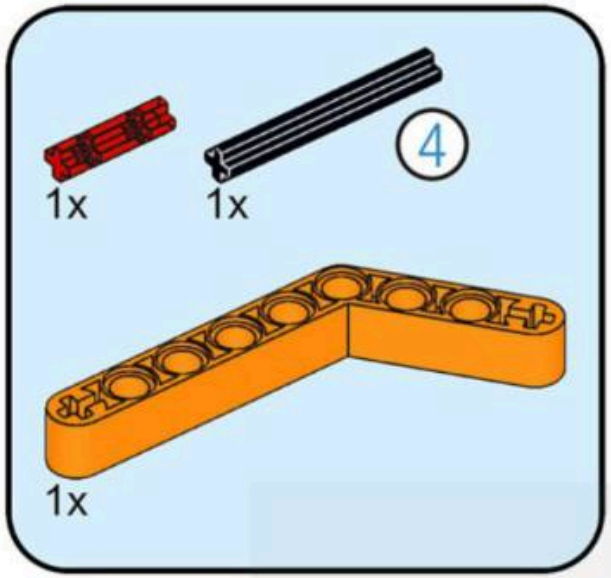
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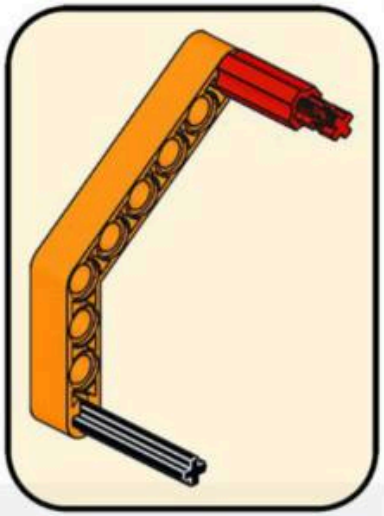


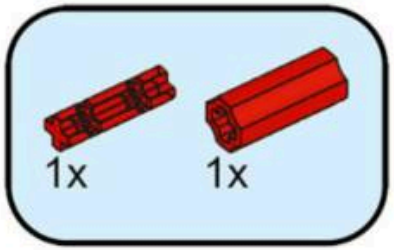
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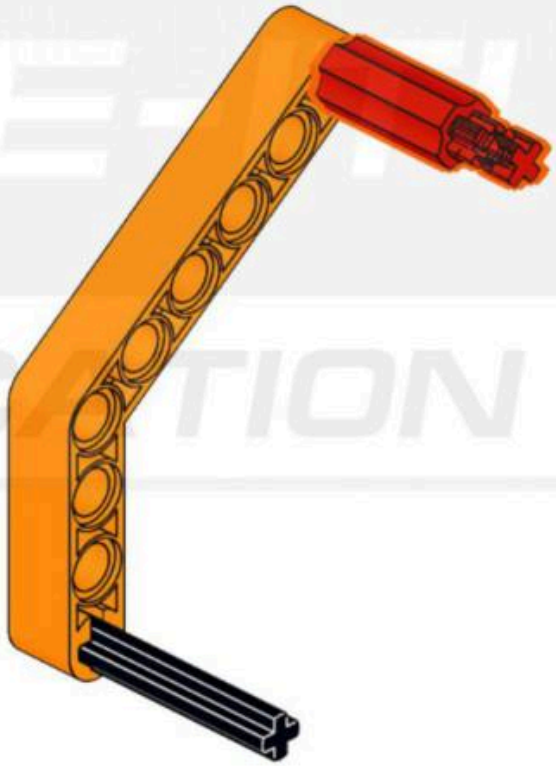
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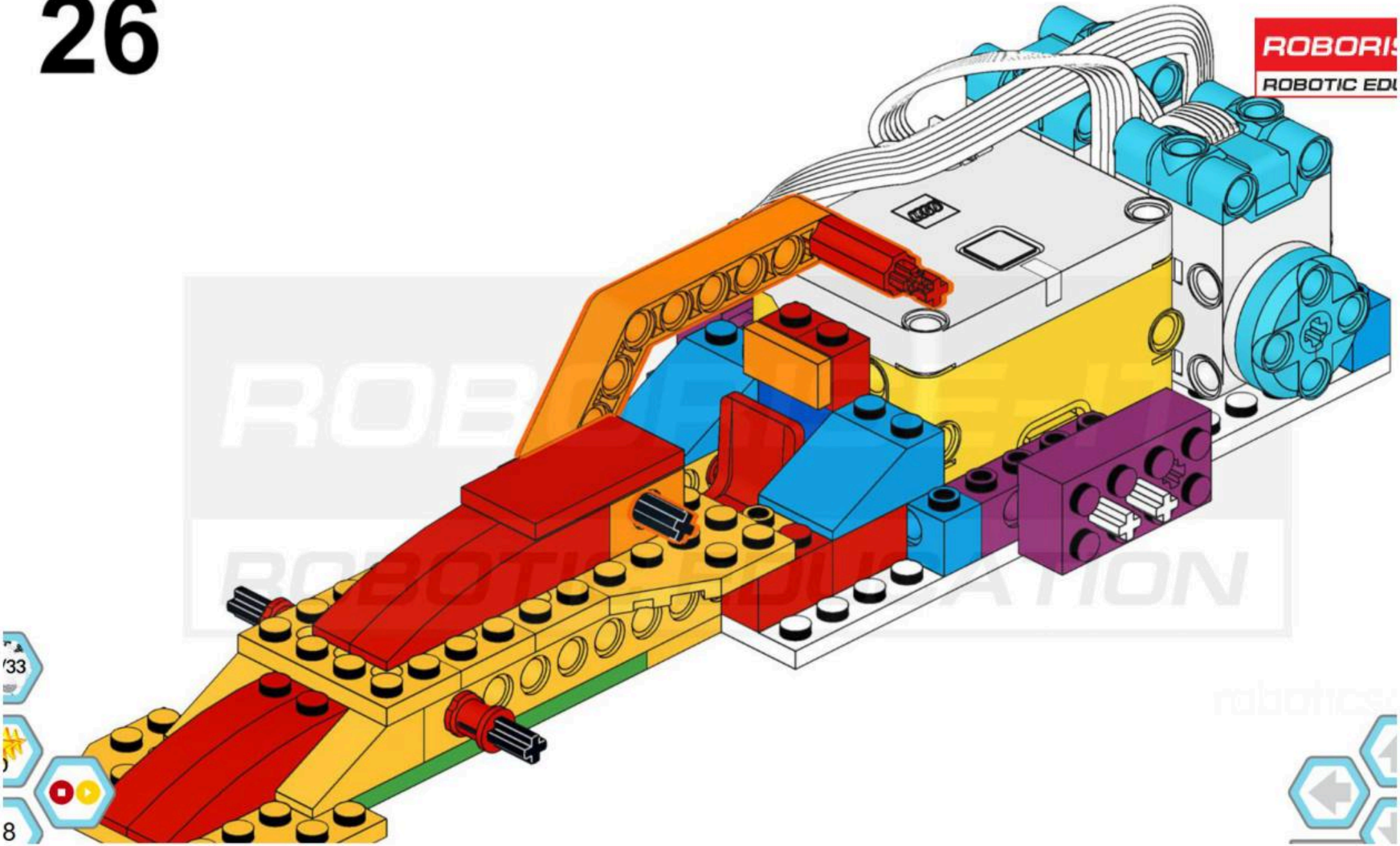


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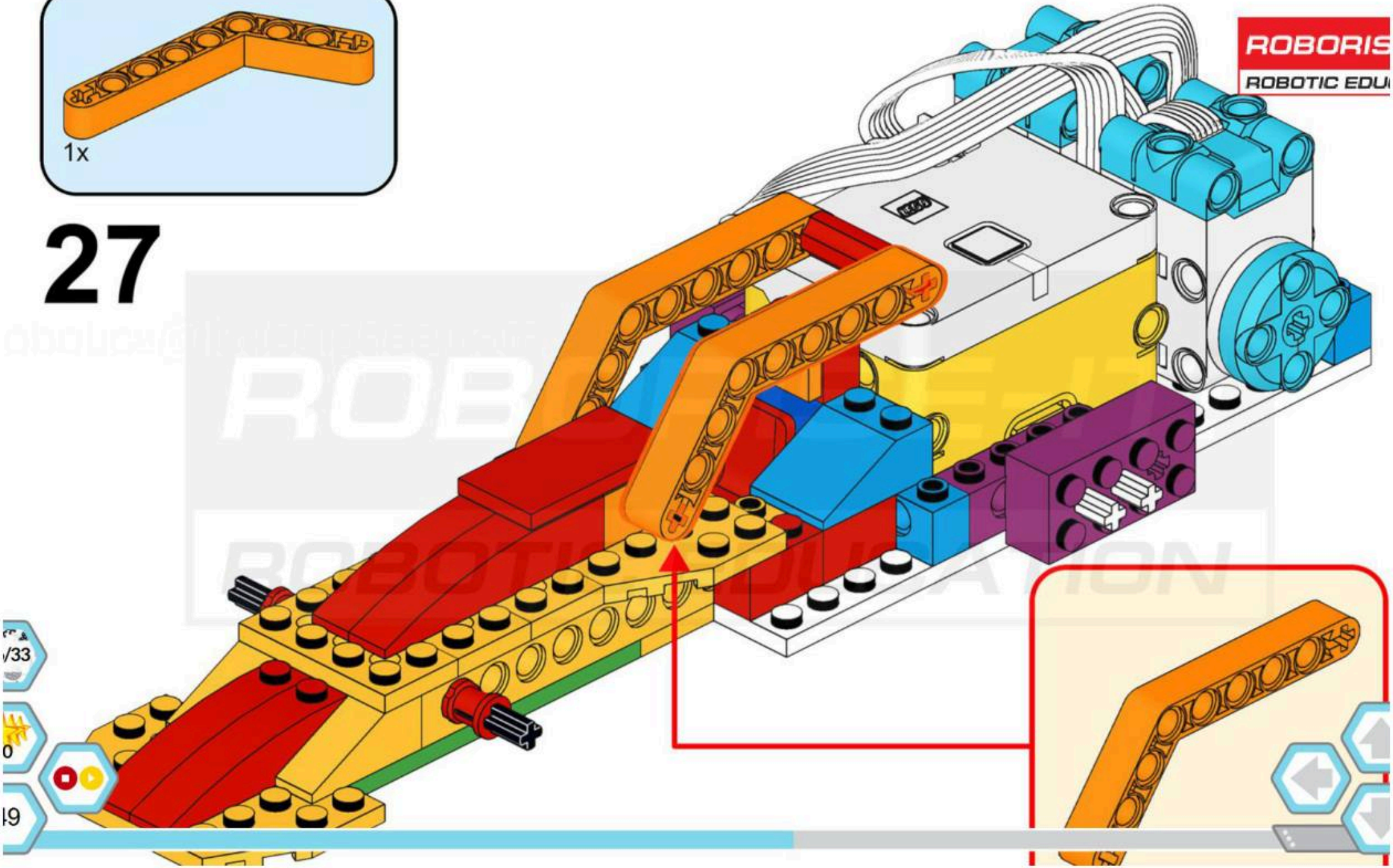


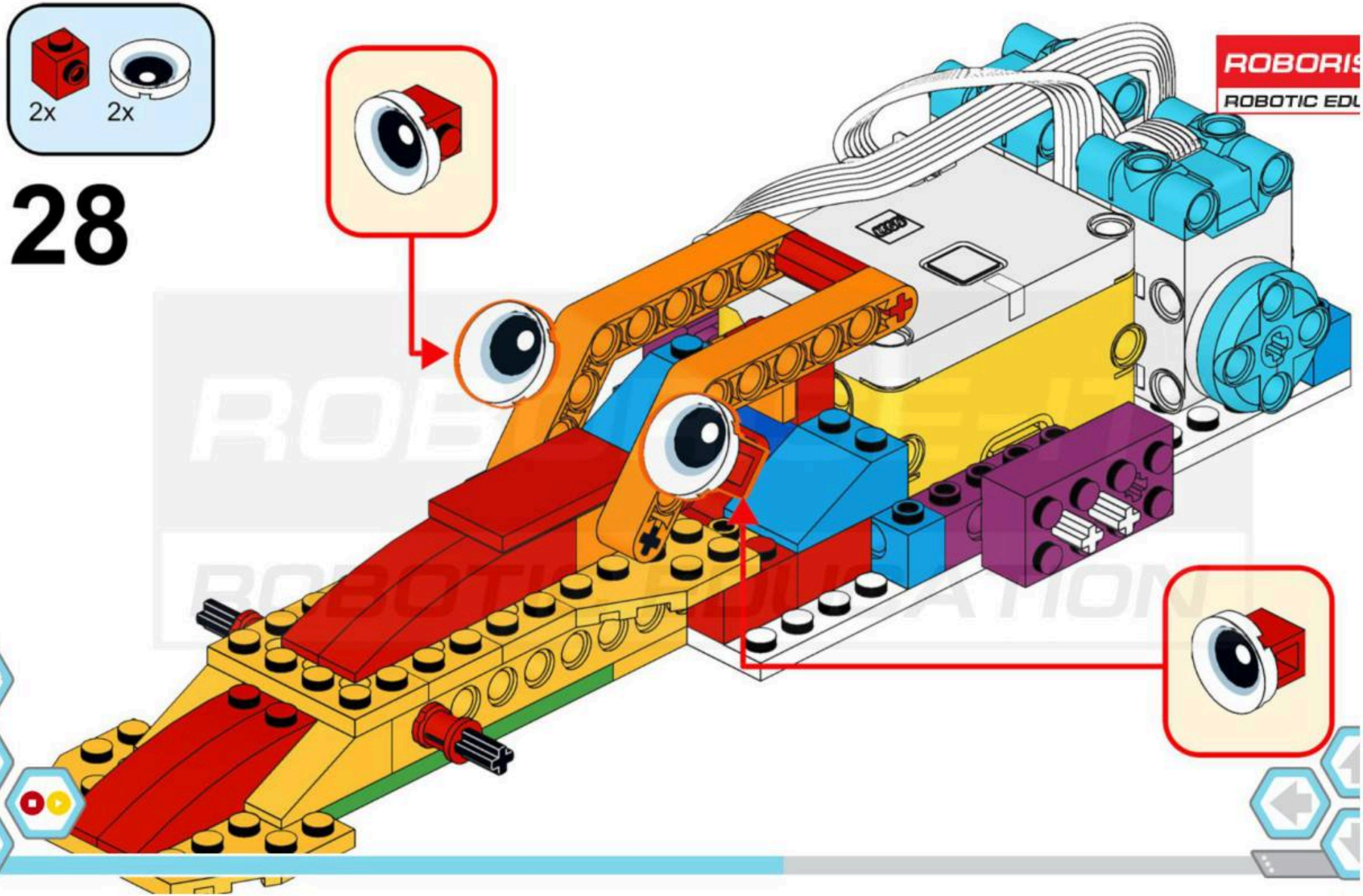
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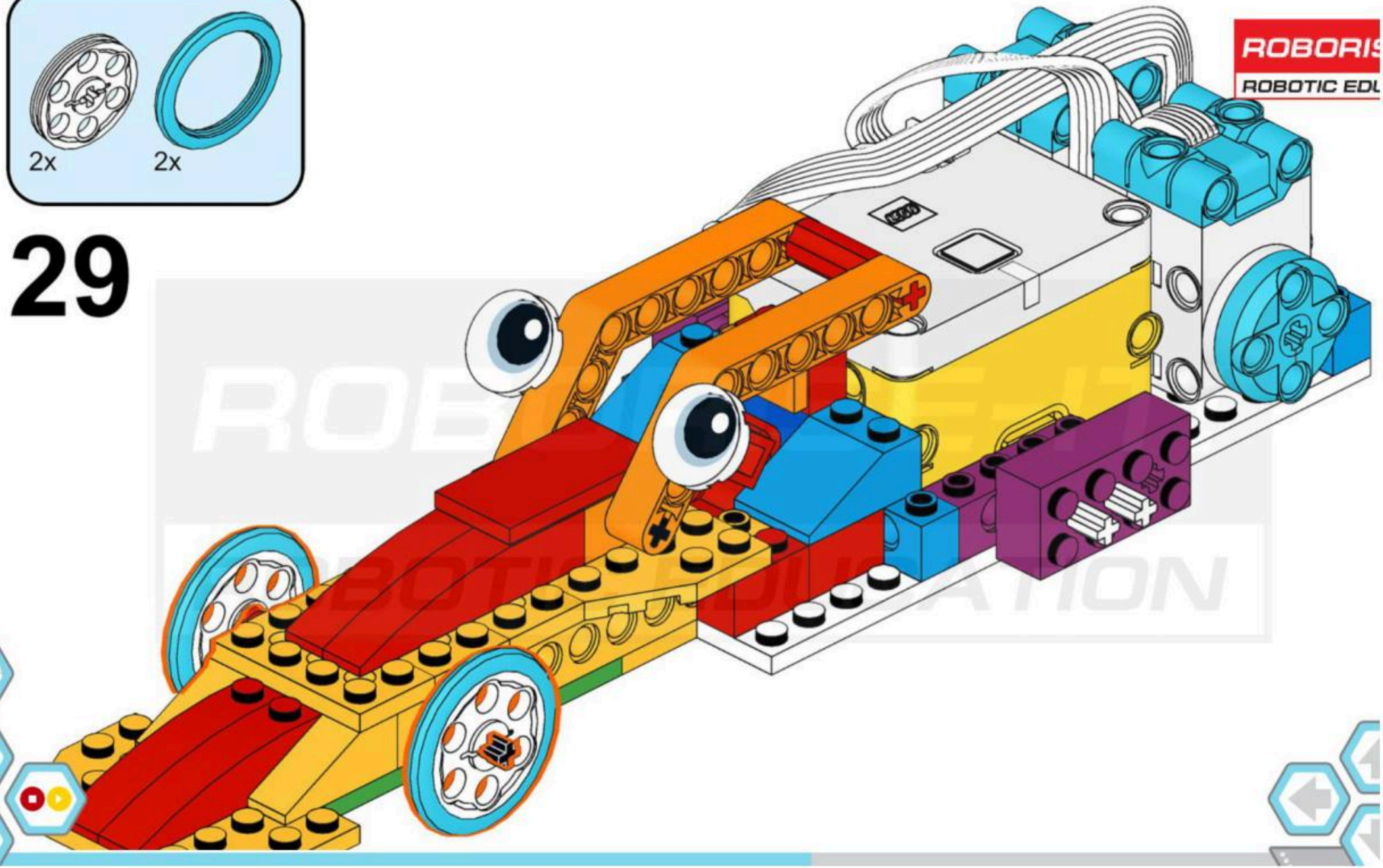


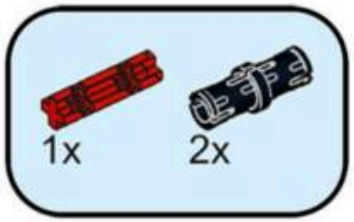
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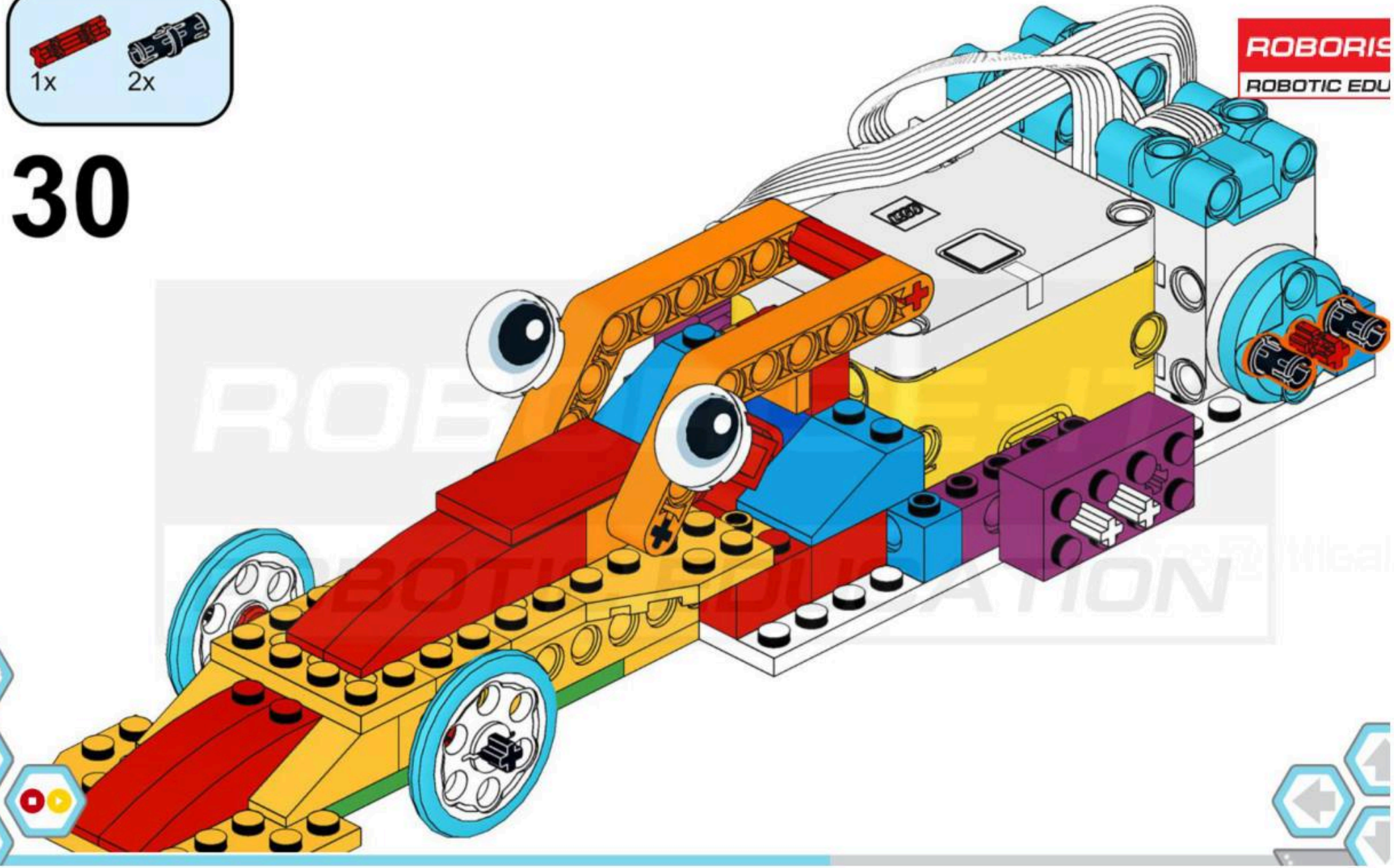

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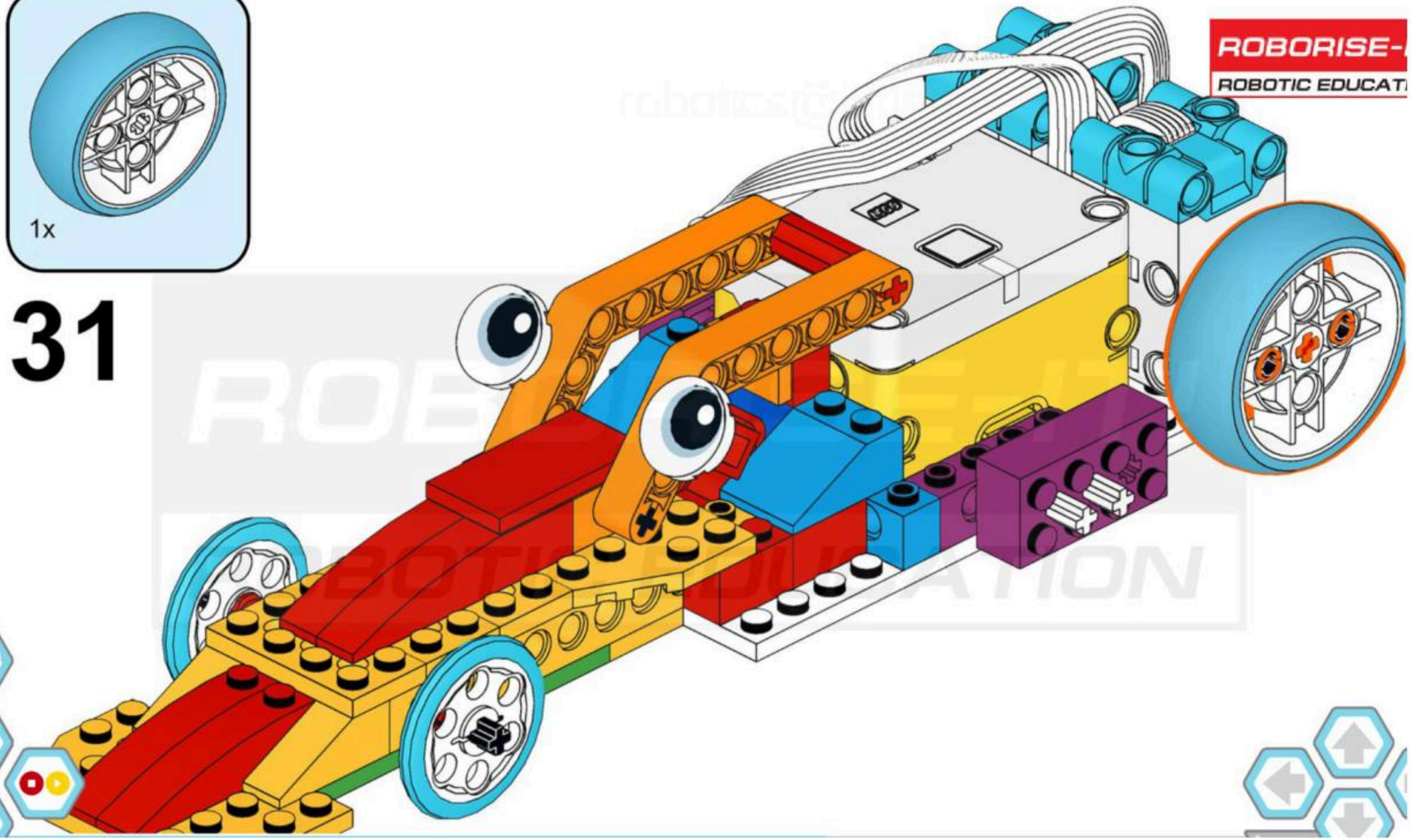


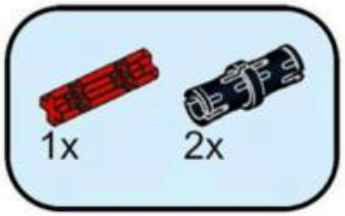


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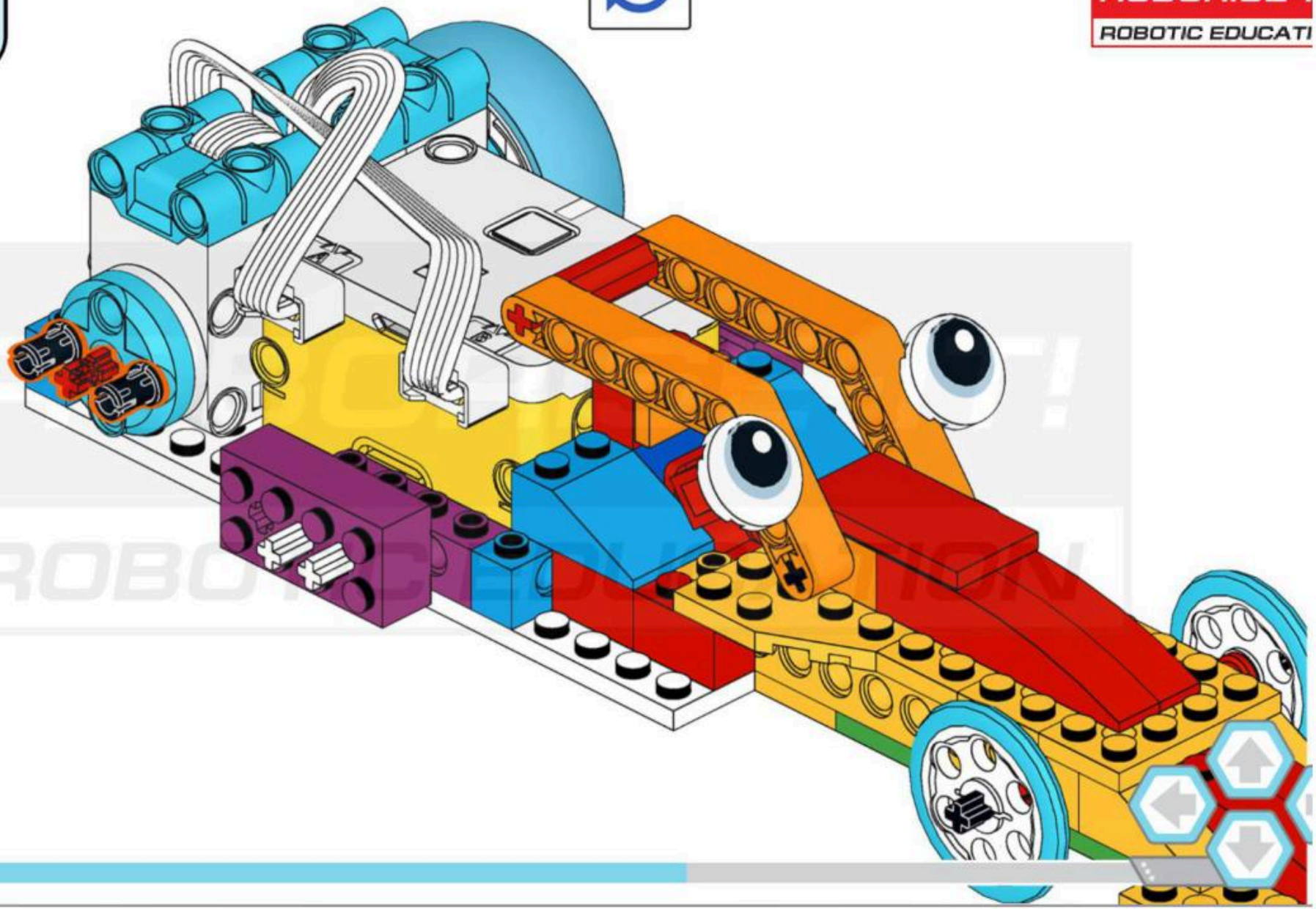


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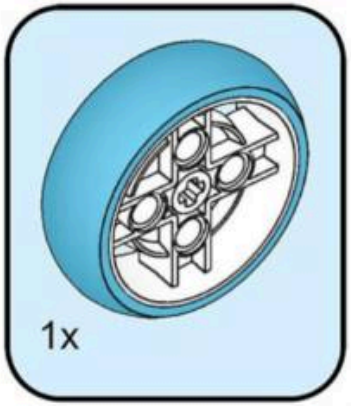




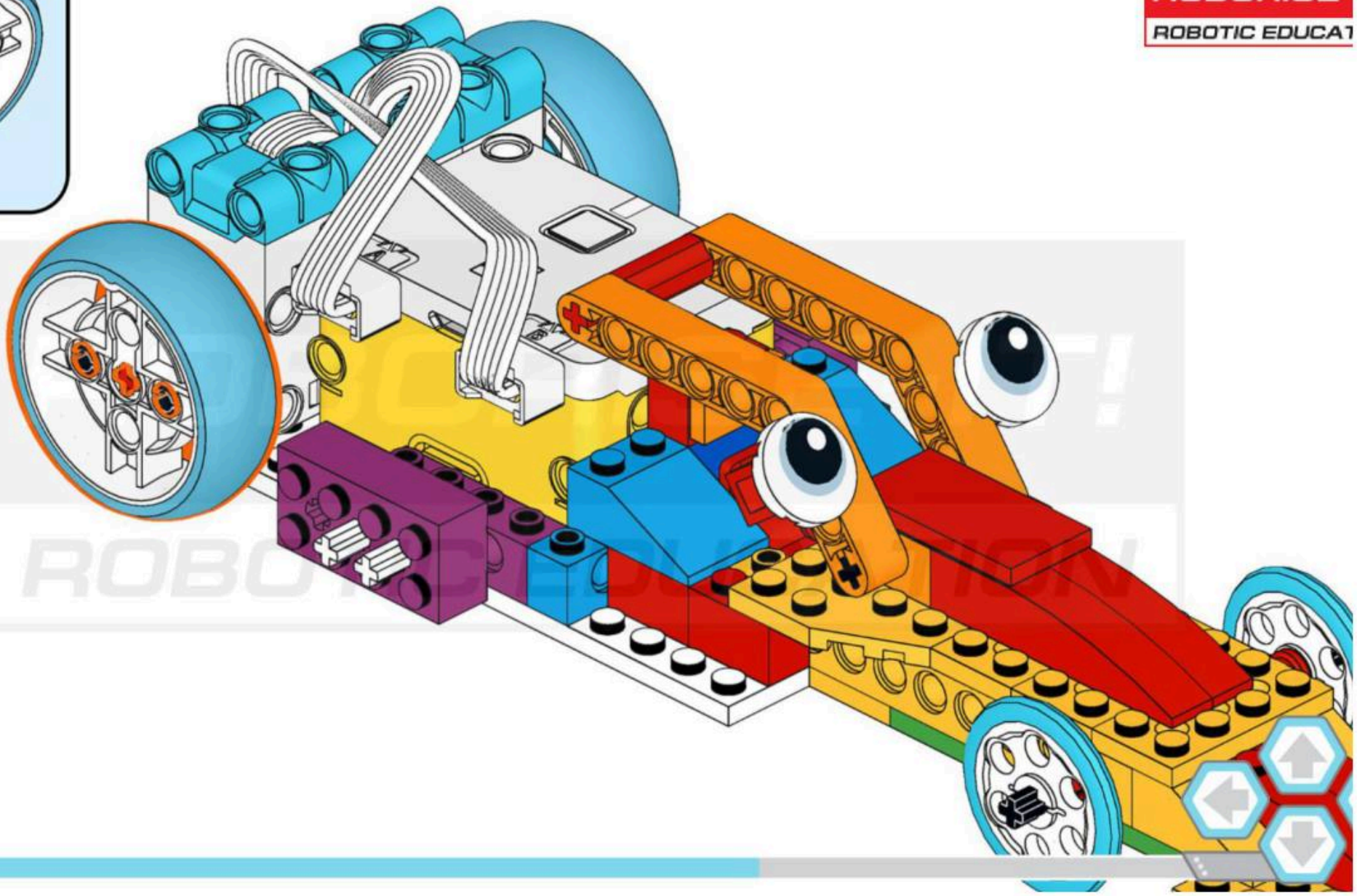
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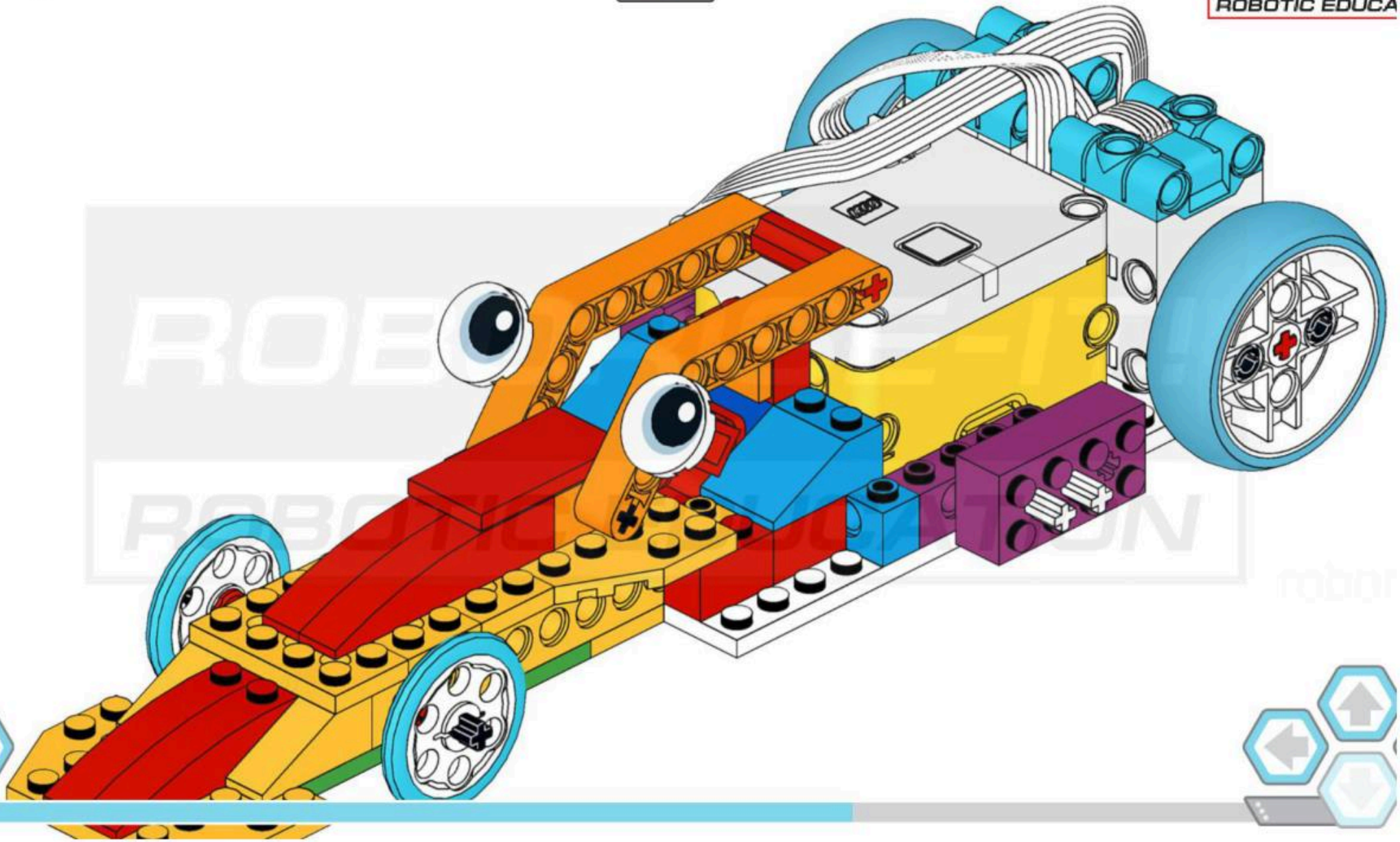


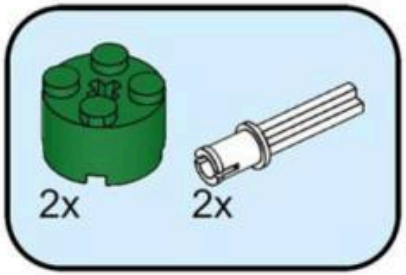
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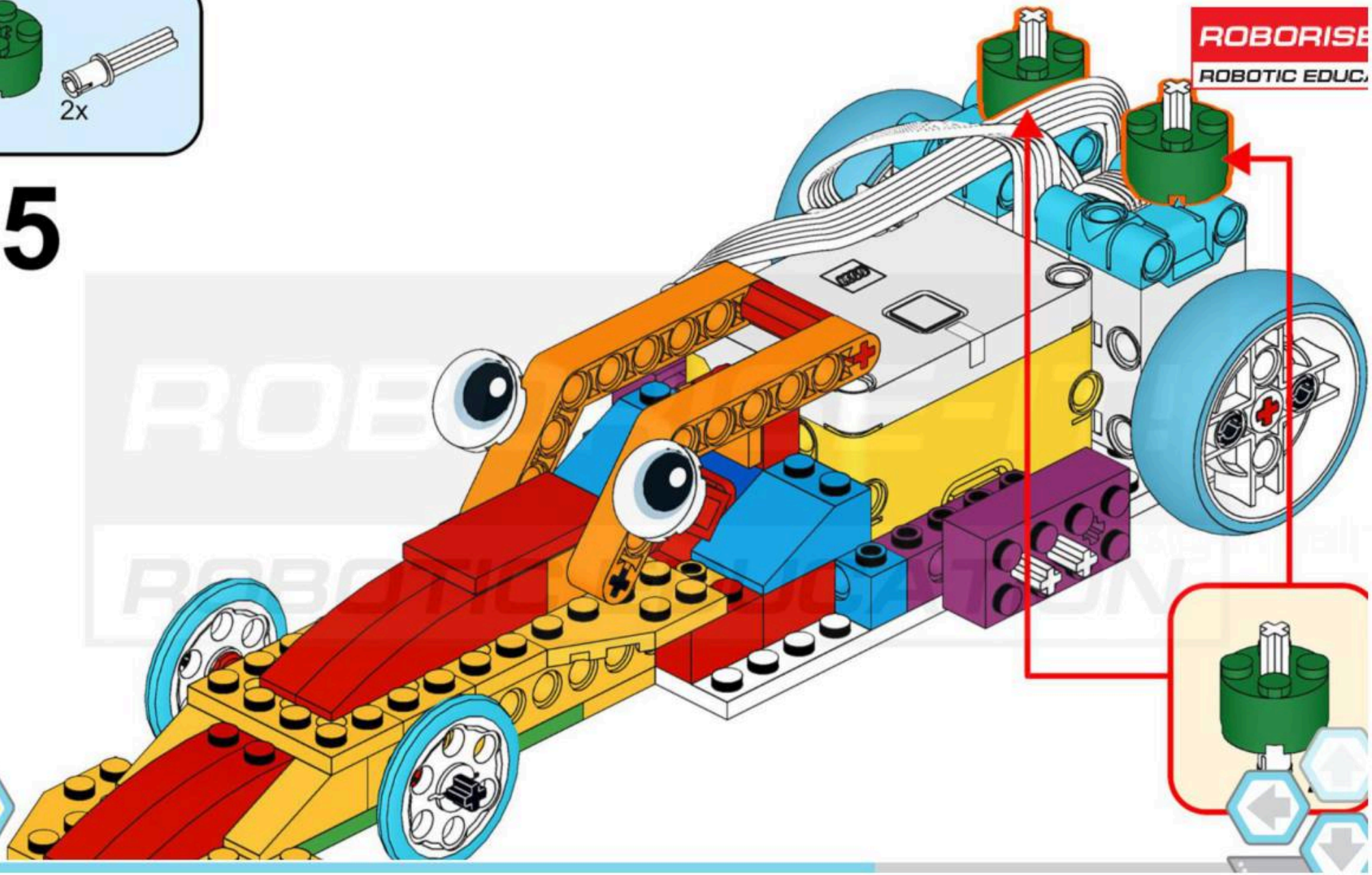
ROBORISE-
ROBOTIC EDUCAT

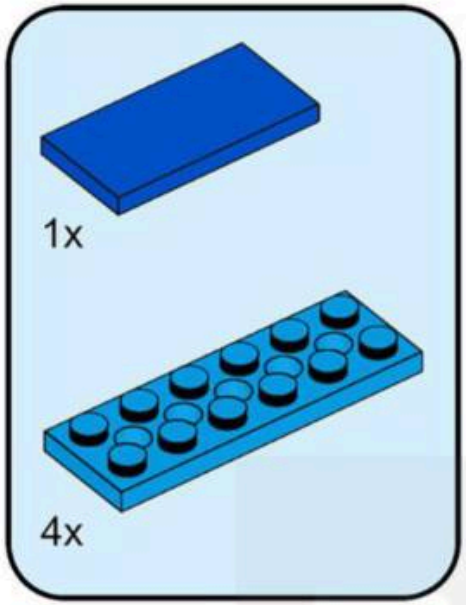
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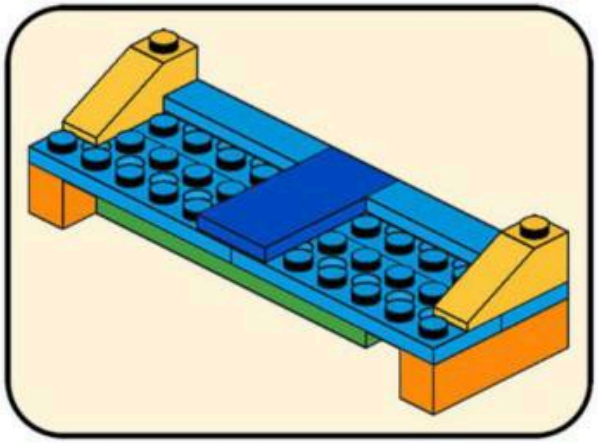




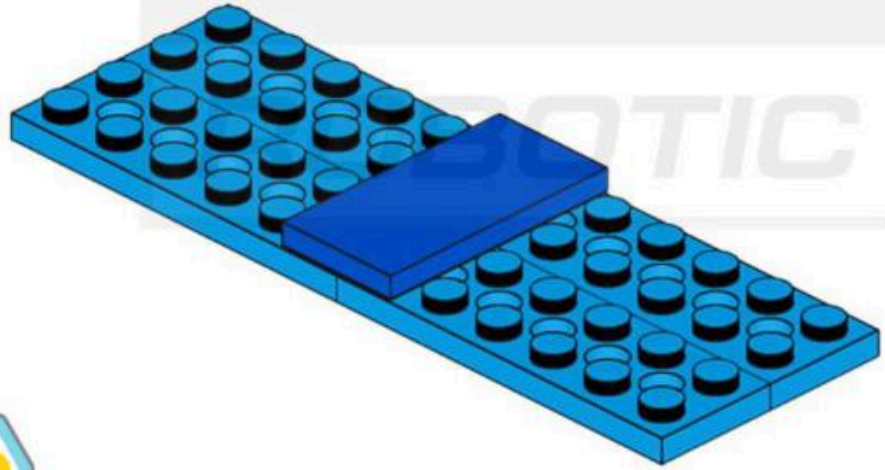
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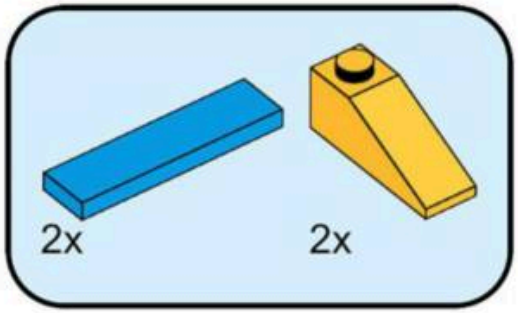




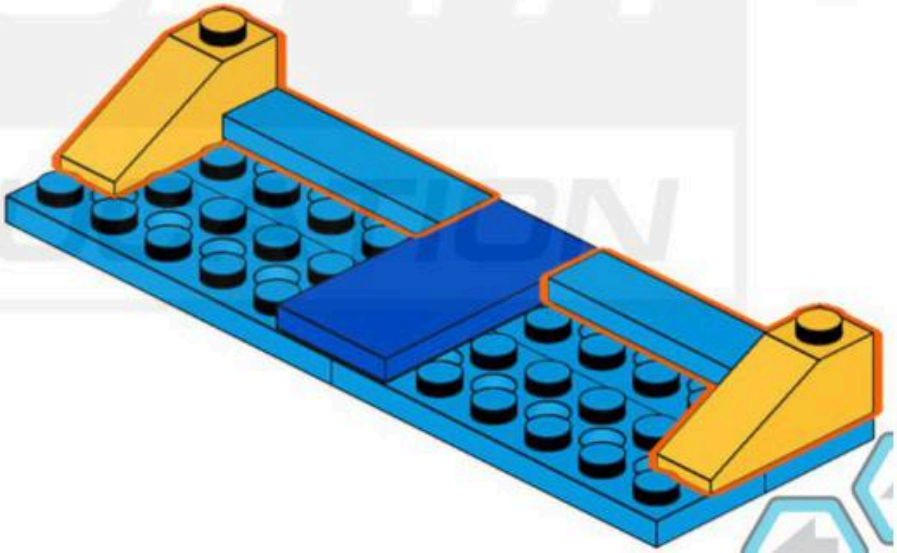


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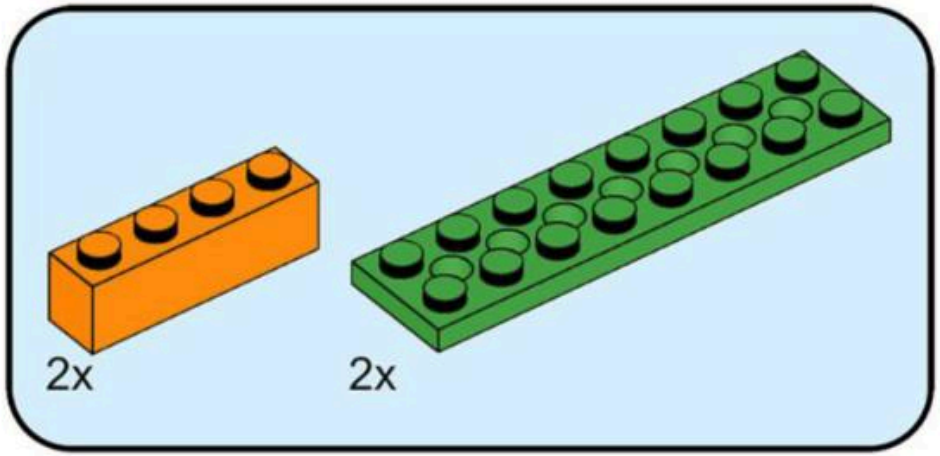




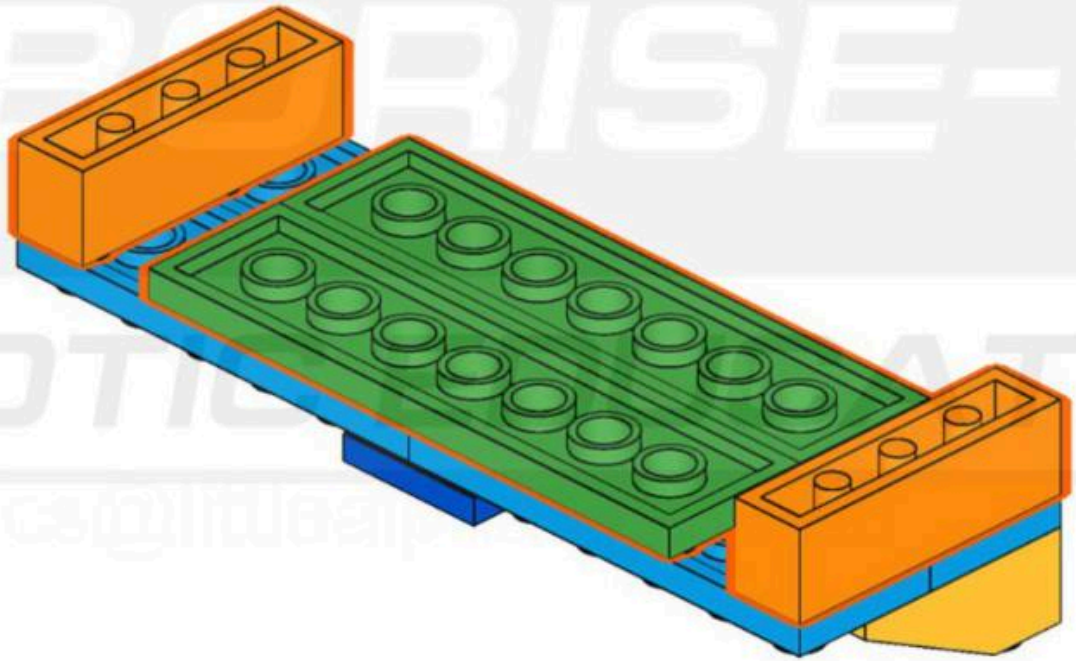
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ROBORISE-IT!
ROBOTIC EDUCATION

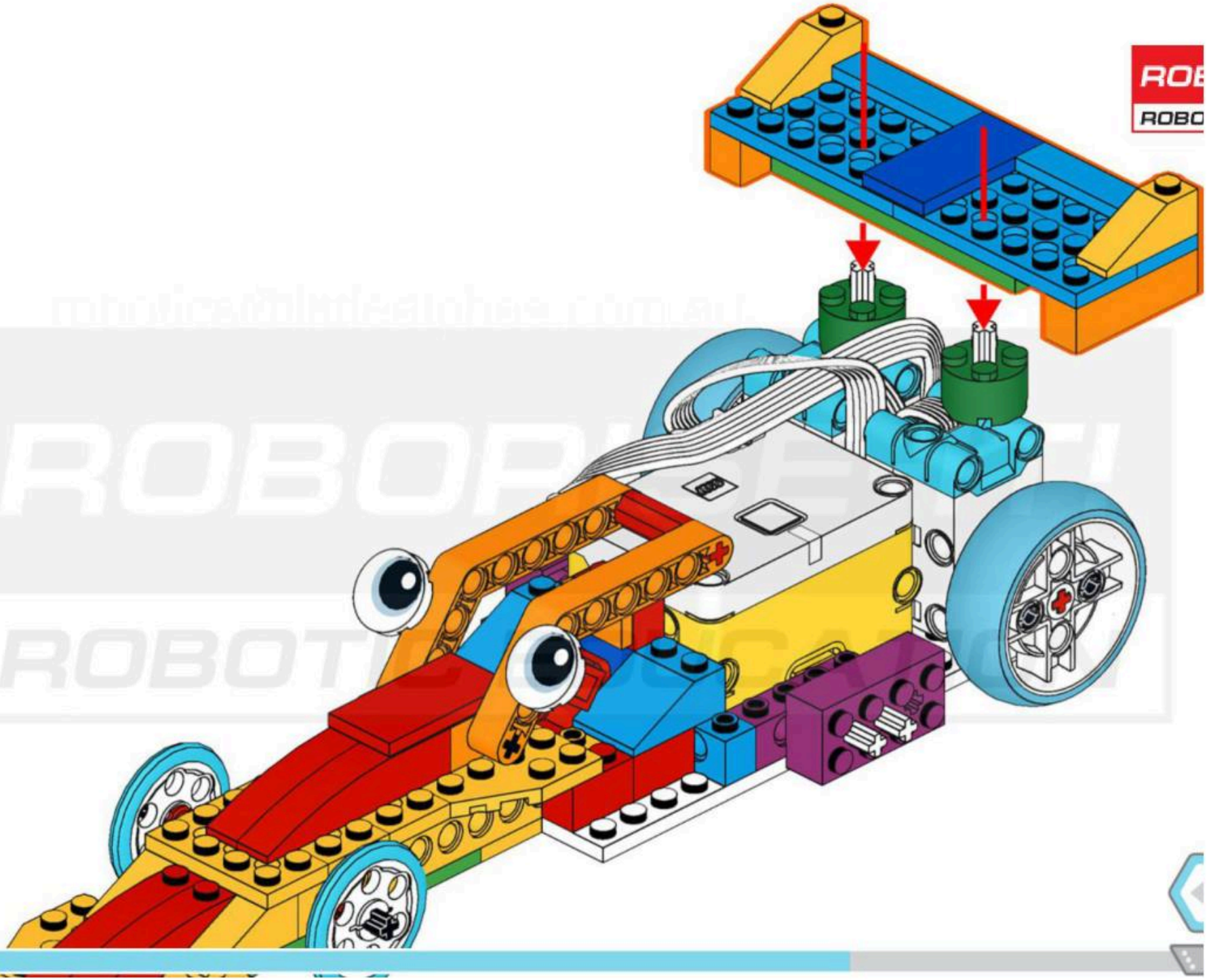


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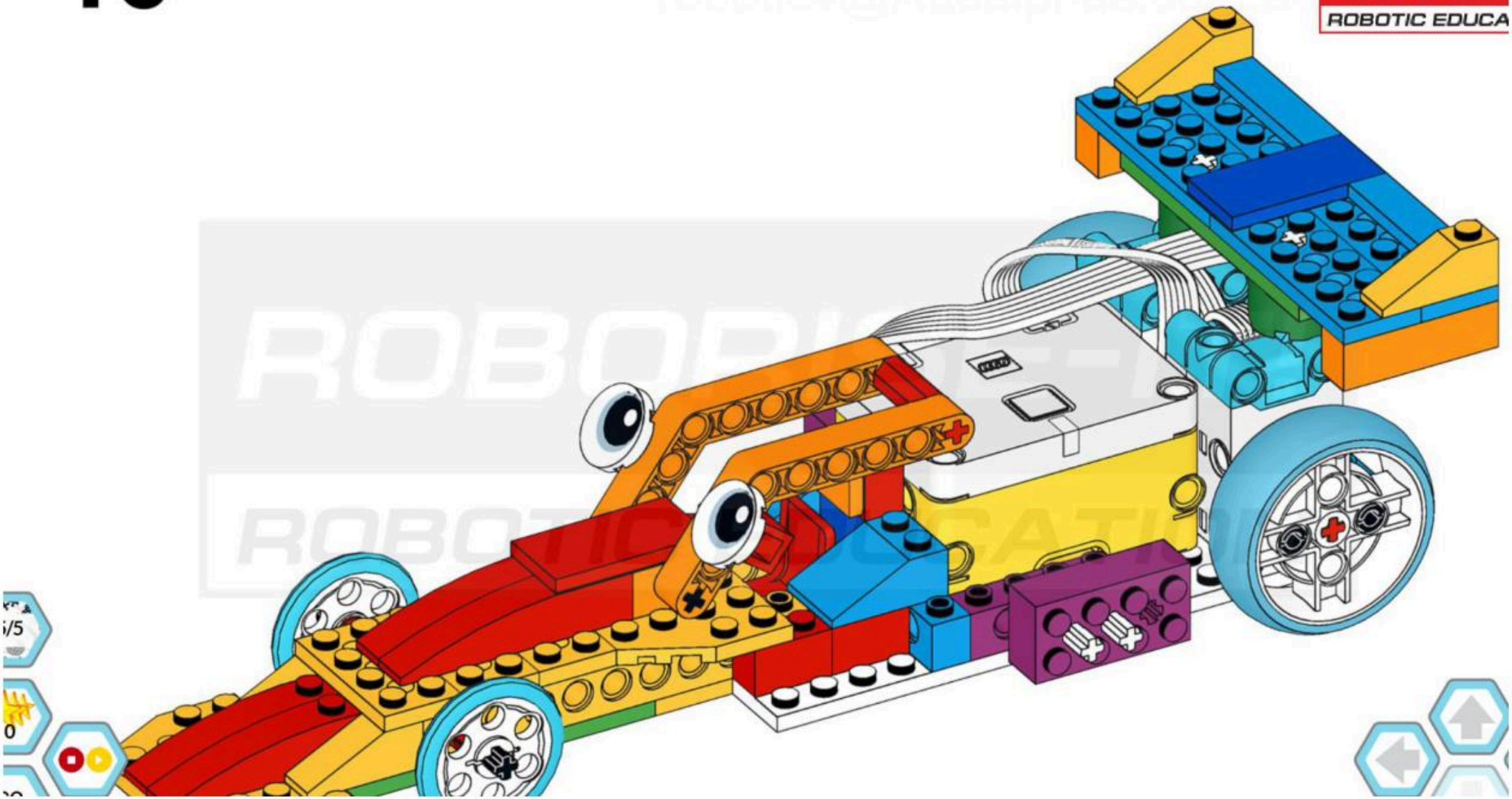


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ROBOTICSBRISE-I

39

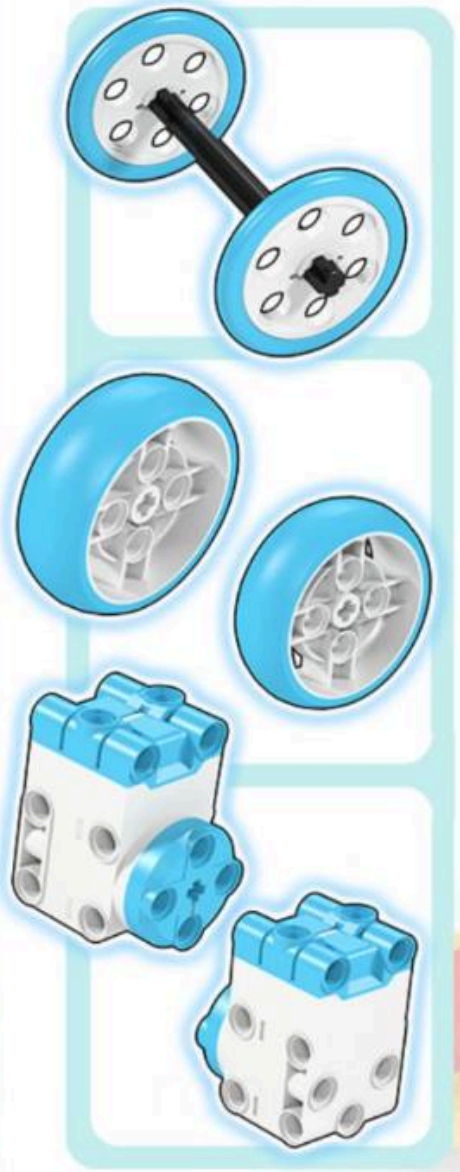


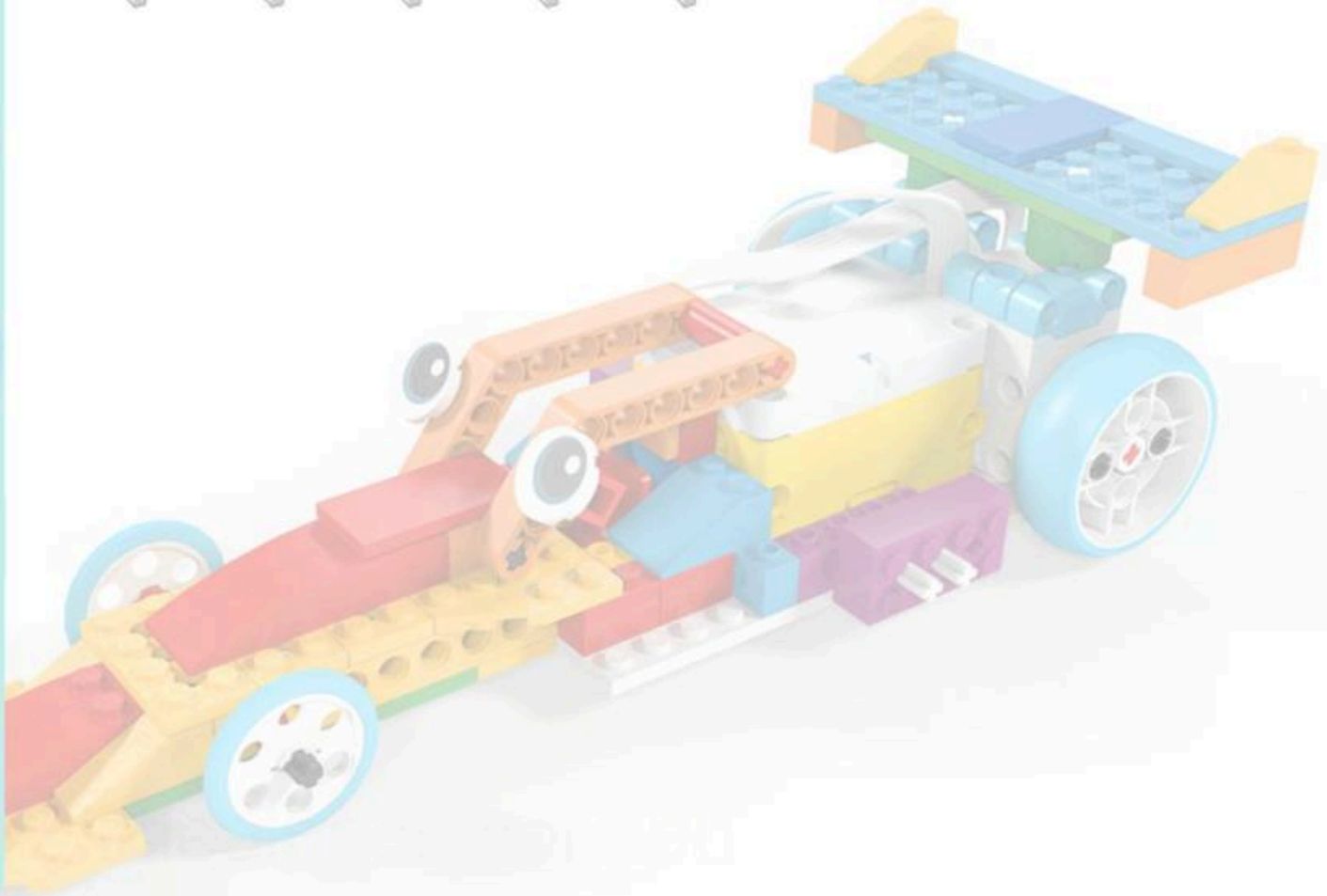
40



Task

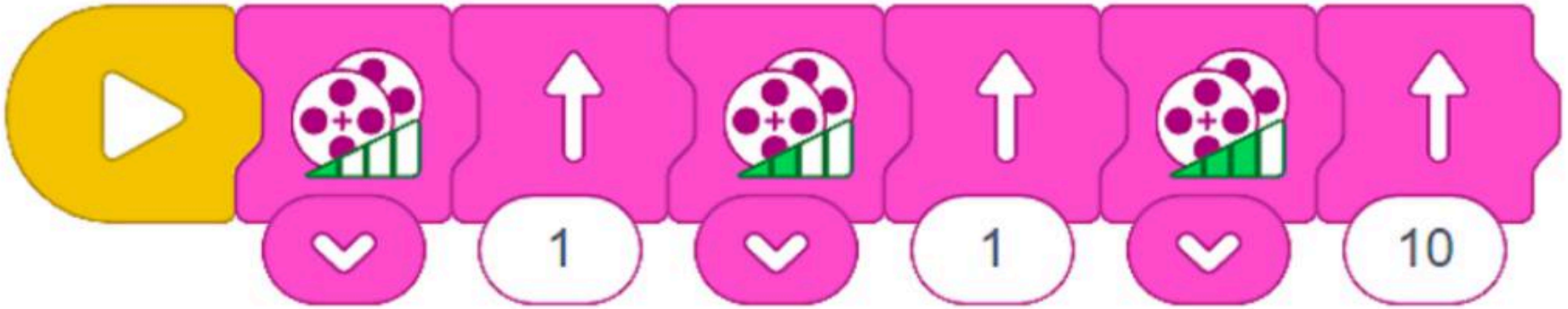
Place the parts of the robot in the correct places





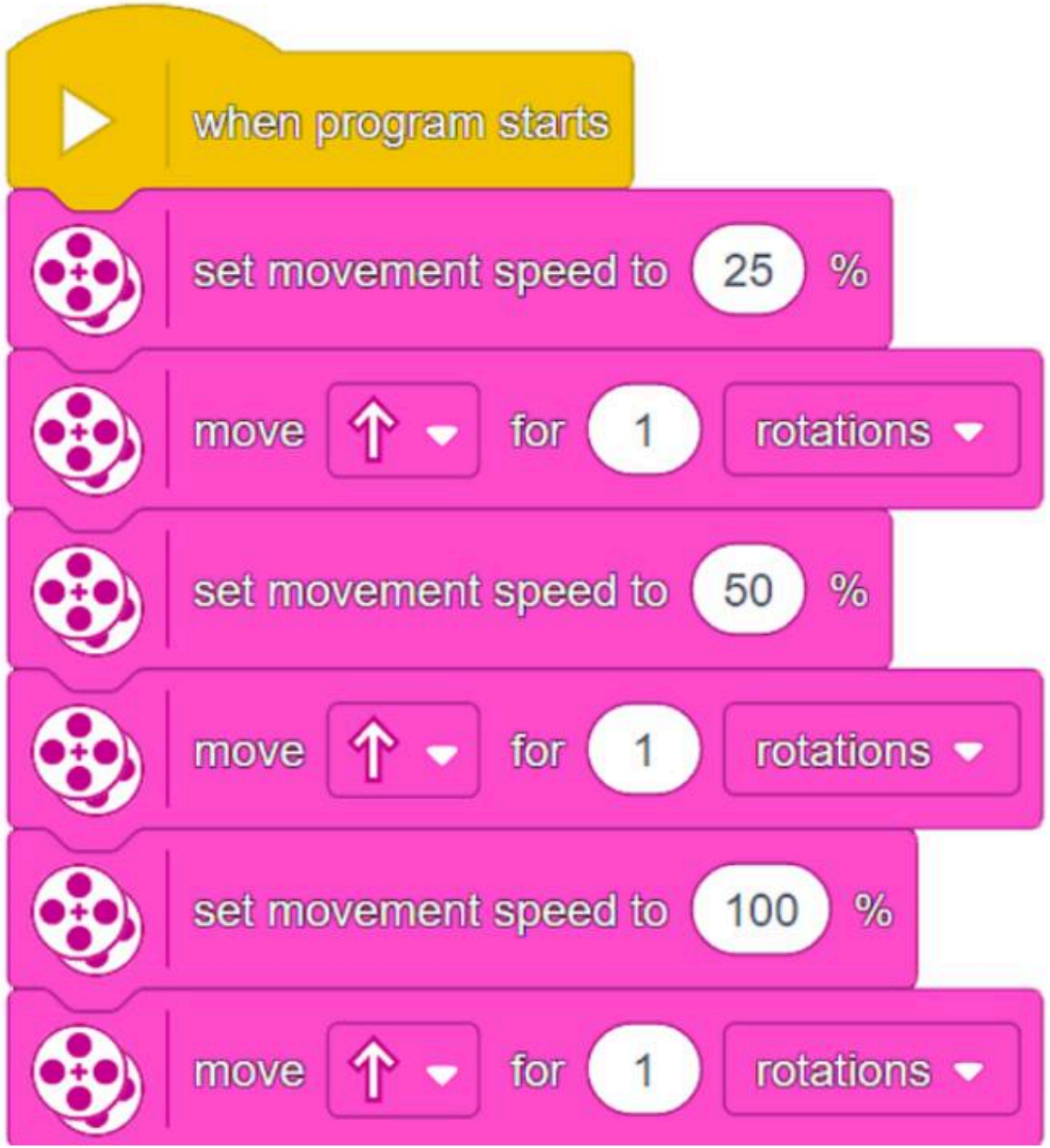
Use motion blocks to program the robot to accelerate:

In ICON Blocks mode, the program looks like this:



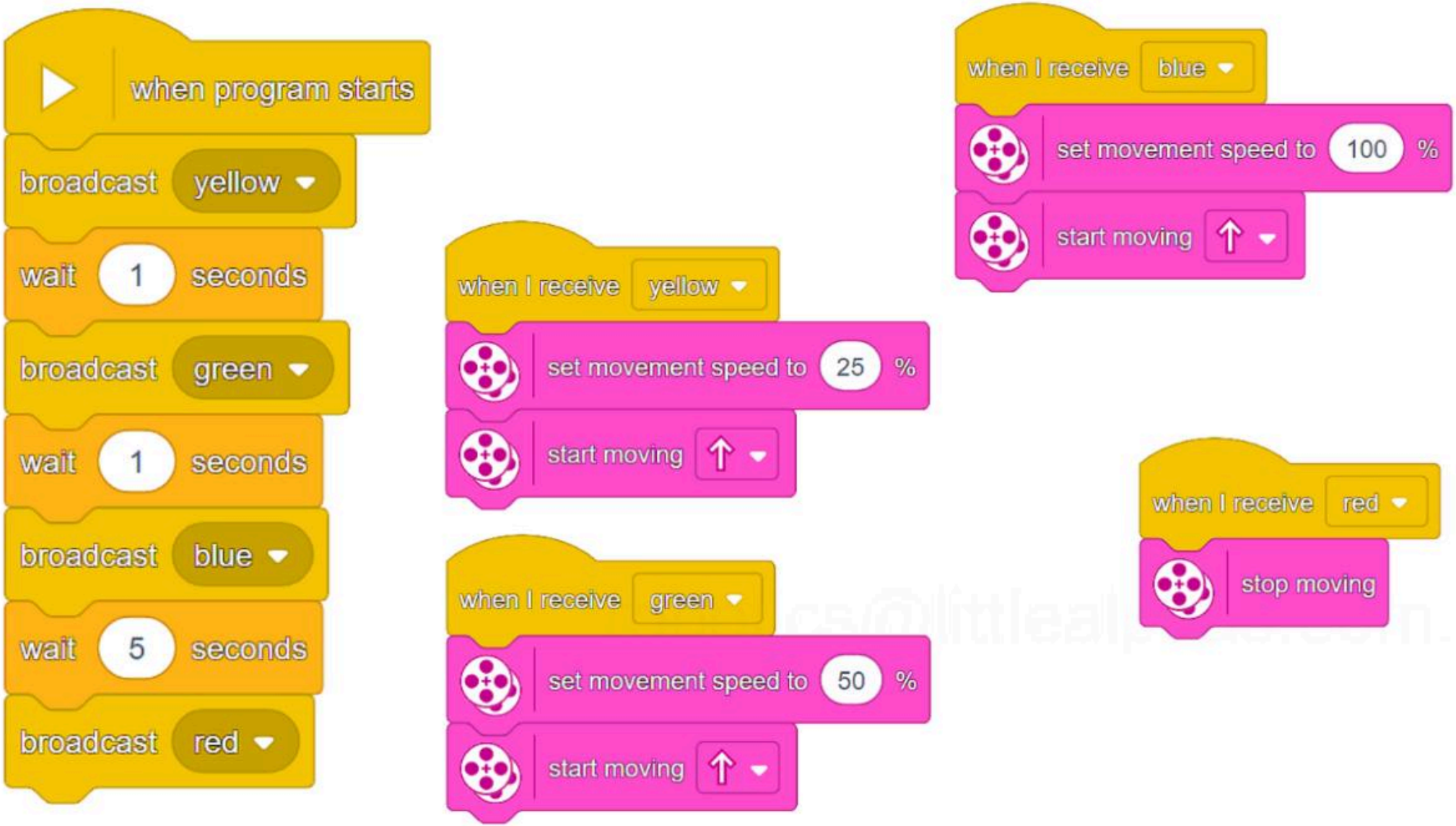
Use motion blocks to program the robot to accelerate:

In Word Blocks mode, the program looks like this:



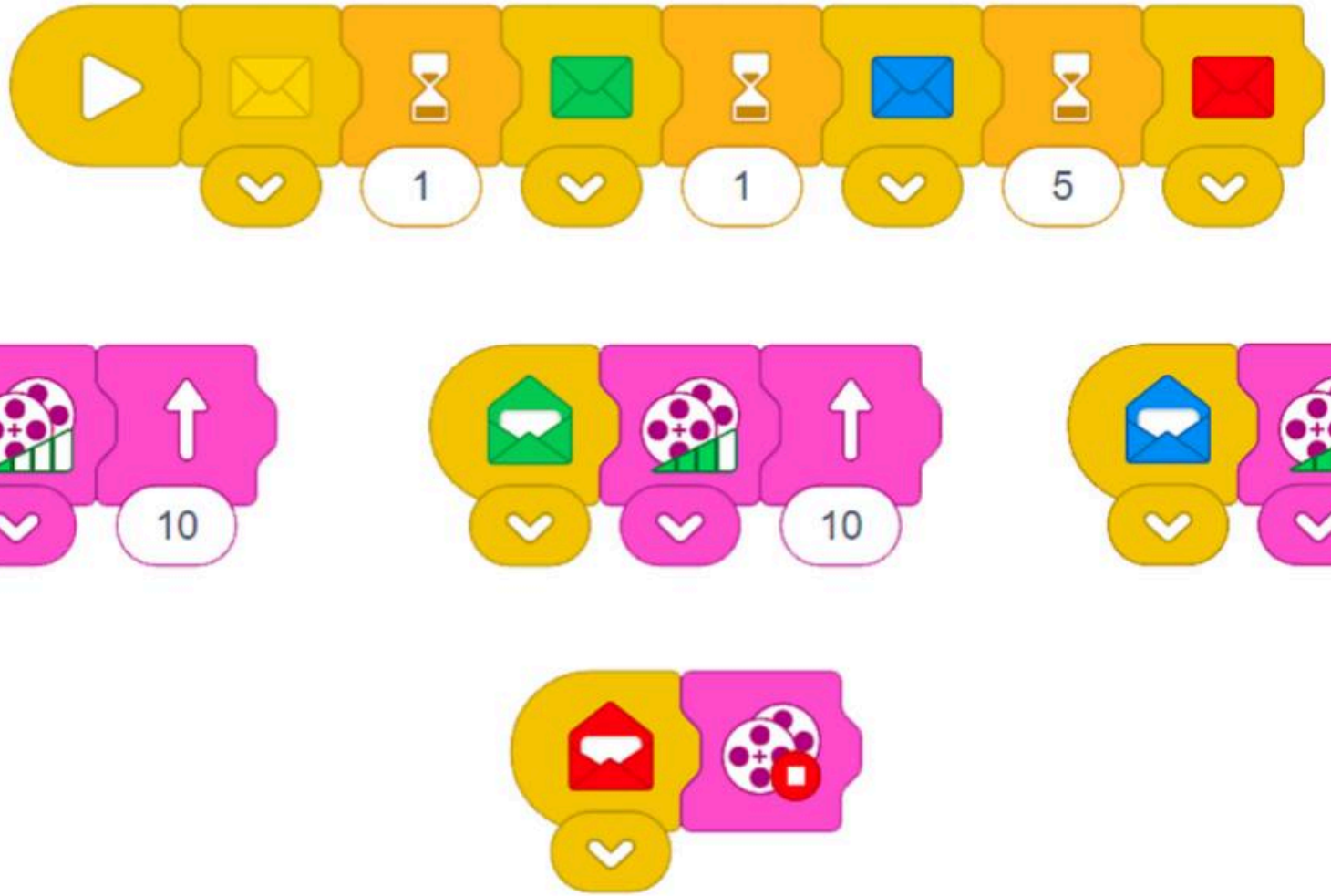
Use sending messages to complete the task.

In Word Blocks mode, the program looks like this:



Use sending messages to complete the task.

In ICON Blocks mode, the program looks like this:



Replace the unit that starts the program and adjust the trigger level (the volume at which the program will start)



Add LED color changing to your programs:

In Word Blocks mode, the program looks like this:



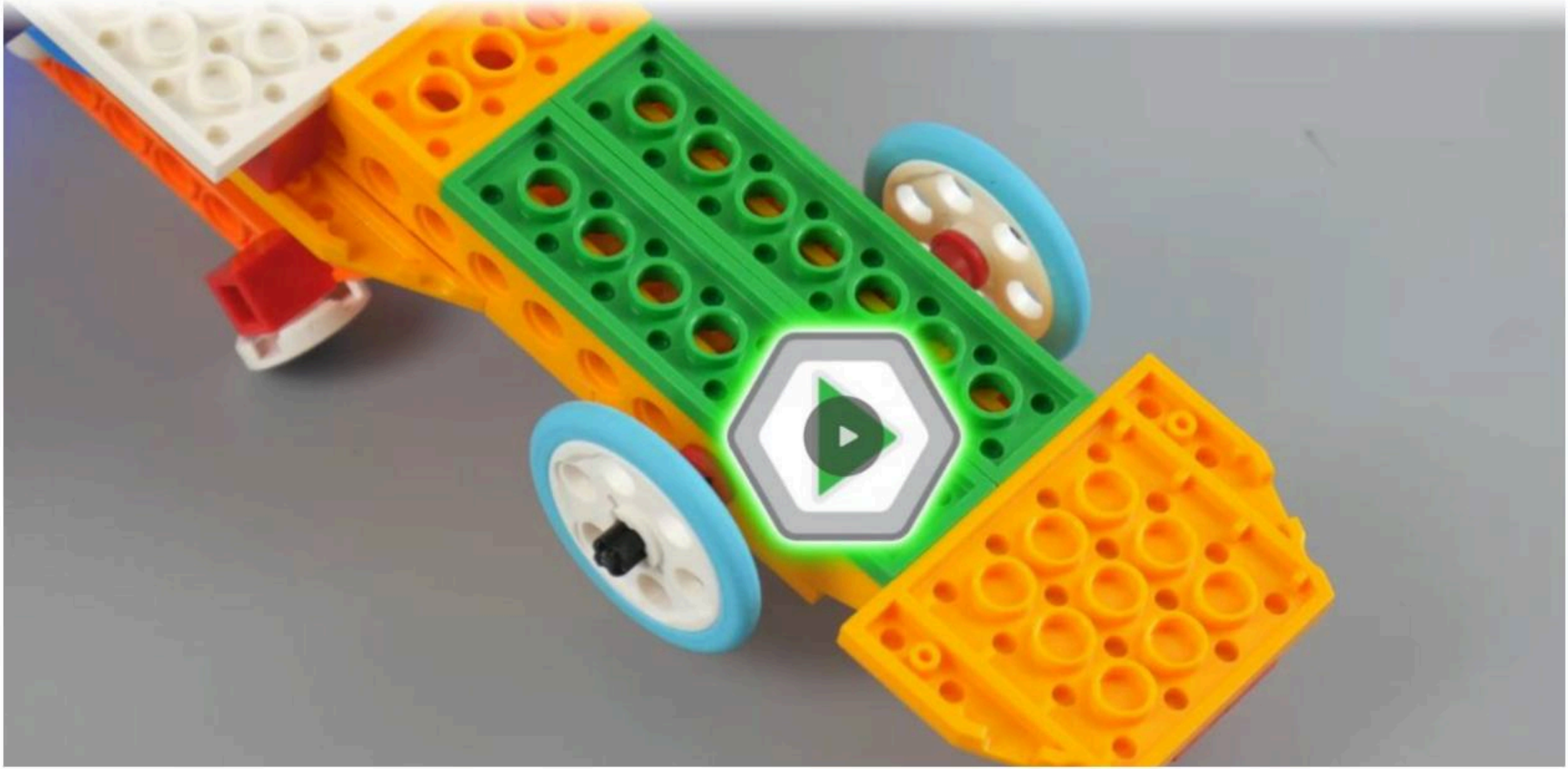






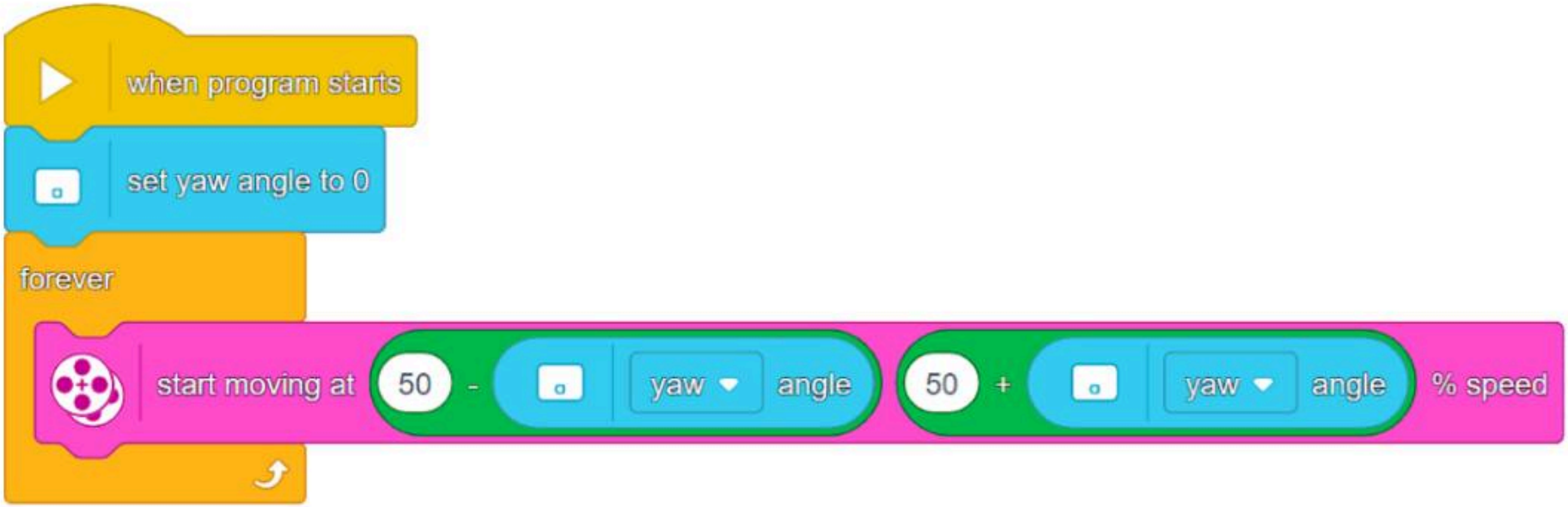
As you know, the robot hub has a gyroscope that can be used to track the robot's deviation from a given position. Program the robot to maintain a given direction even in the presence of external factors affecting its movement

At the same time, the front wheels of the robot have a fairly high grip on the surface, and if the tires are not removed, the testing may not be so revealing. Therefore, to test the program, remove the tires from the front wheels



Program and test your robot.

In Word Blocks mode, the program will look like this.



The idea is that the deviation from the given direction causes the power of the left and right motors to change. When you deviate to the left, the power of the left motor will increase and the power of the right motor will decrease, and vice versa. This is how the robot movement will be corrected.

Question

The dragster has two wings, the front and the rear.
What is the direction of the forces they produce?



Up

Down

Opposite directions

Analyze the programs and calculate the approximate duration of the robot's movement. Select the right answer.









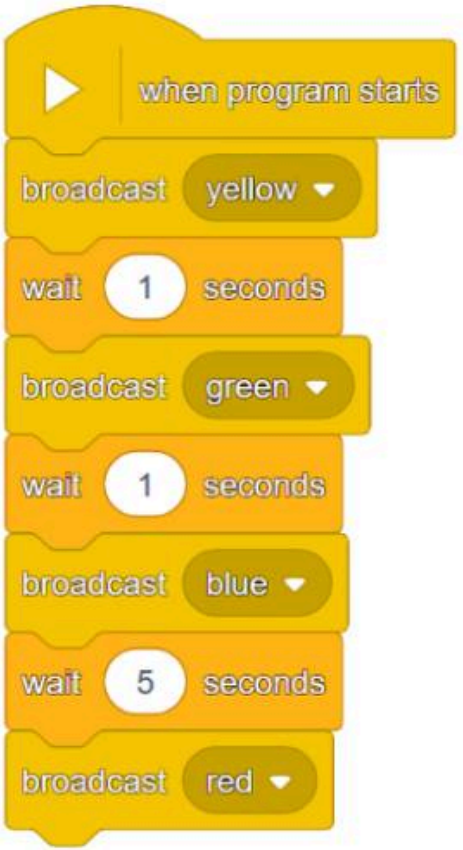












1 s

5 s

6 s

7 s

Your achievements

Total: 0

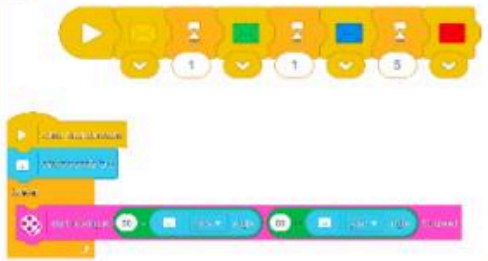
1



2



3



4

